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Young Children's Spontaneous Comprehension of Symbol-Referent Relationships in the Graphic Domain

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Abstract:	Young children's communicative skills in language and gesture are well documented, yet much less is known about their ability to interpret graphical symbols. Across three coordinated studies using identical procedures, we examined children's spontaneous comprehension of unfamiliar graphic symbols in a simple object-choice task spanning a broad range of symbol–referent relationships. Each study included four conditions in which reference was established through direct representation, part–whole relations, analogies in size, number, and form, or gestalt principles such as orientation, alignment, and figure–ground organization. Participants were 304 German-speaking children continuously sampled from age 3 to 7 (Study 1, N = 106, 51 female; Study 2, N = 99, 49 female; Study 3, N = 99, 55 female). Children reliably used direct representations to identify target objects before age 3, whereas group-level success in the more abstract conditions emerged later but consistently before school entry. Converging evidence across conditions involving abstract or conceptual correspondences indicates a major shift in symbolic competence around age 4. Together, these studies trace children's developing ability to understand graphical reference across a continuum from iconic to abstract and offer one of the most systematic investigations to date of how young children derive meaning from unfamiliar graphic symbols.

SCHOLARONE MANUSCRIPTS

Response to Reviewers

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in the Graphic Domain

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We are very thankful for the reviewers' comments and think the manuscript has highly benefited from their suggestions and questions. Below we provide a detailed overview of all points that were raised and how we responded. In order to make the review process more convenient, all points have been numbered (e.g. R2.4 might be read as *reviewer 2, point 4;* also see E.1. for points from the editor). All comments by reviewers and editors are in italics.

Editor's Notes

Overall Evaluation

All three reviewers brought up the theoretical clarity of the introduction, and would like to see revision. I agree. In its present form, the introduction does not present a clear, coherent description of the theoretical motivation for the present research. In some places, this simply involves being more precise with your choice of phrasing. In other places, however, this will involve significant conceptual reorganization and rewriting to ensure that you are motivating only the ideas that are necessary for the present study. I will encourage you to look carefully at the main points made by Reviewer 1, the major comment under the heading, "theoretical" by Reviewer 2, and points 1-3 by Reviewer 3.

E1.1

To offer you a concrete suggestion in your revising, the introduction is also a little long, and could be shortened by 500-1000 words. If you try to revise with that constraint in mind, it might help focus you on the specific phrasings and arguments to clarify your position.

Response:

Thanks! To address this we have reworked the introduction. Rather than going through the literature discussing all symbol-referent relationships tested across the 12 conditions of the three studies, we now start out with a very brief review of the development of communicative competences and then provide a detailed overview of the steps required in solving symbolic object-choice tasks. This allows us to introduce all relevant terminology and discuss the inferences children need to make when solving these tasks, separating pragmatic and analogical reasoning and reviewing results for both strands of literature separately. This in turn helps to highlight the contribution of our work. To focus only on ideas that are central to our investigation, we have suppressed references to children's production of drawings considerably shorten the discussion of stimulus dimensions.

E.1.2

Further, I want to remind you that not all readers will be familiar with the Bayesian analyses that you present. For the purposes of readability, you might want to step the reader through your analysis a little more clearly as well, to ensure that all readers are following the arguments you are making.

Response:

We have rewritten the paragraphs outlining the analyses. In addition to providing all information necessary for reproducing the analyses, we now also describe why the respective steps and decisions were made. While this may be a bit verbose for readers with expertise in Bayesian modelling, it should be very convenient for readers that are less familiar with this analytic approach:

> *To evaluate how children's ability to interpret cues varies by age and condition, we employed Bayesian Logistic Generalized Linear Mixed Models (GLMMs). This mode of analysis is appropriate as it allows to estimate the probability of success while accounting for uncertainty and individual differences, and because it allows mapping linear predictors to a probability between 0 and 1 via a link function. Our analyses modeled children's binary choices (correct vs. incorrect) using fixed effects for condition, age as well as their interaction as primary predictors. In addition, we included a fixed effect of trial number, to control for learning or fatigue, and a fixed effect of sex. To account for the possibility that some children may be better at the task than others, we included random intercepts for each participant. We also included random slopes for trial number, allowing potential effects of learning or fatigue to vary from child to child. To facilitate processing and interpretability, age and trial number were standardized to a natural mean of 0 prior to modeling. The full model notation was correct ~ condition*z.age + z.trial + sex + (z.trial | subid). In line with best-practice conventions, we evaluated the relevance of age and condition for children's performance prior to interpreting coefficients. For this, the full model outlined above was compared with a reduced model lacking the interaction of age and condition. To assess the fit and relative explanatory power of either model, we used Widely Applicable Information Criterion (WAIC) scores and weights (McElreath, 2018) as well as the difference in Expected Log Predictive Density (ELPD) via the function loo_compare (Sivula, Magnusson, Matamoros, & Vehtari, 2020). Across all studies, the preregistered full model proved superior or equally predictive. Model estimates were inspected for the different predictors including their 95% Credible Interval (CrI). In each study, the condition hypothesized as the most simple was set as the reference level within conditions. All models were implemented in Stan (Stan Development Team, n.d.) via the function brm of the package brms (Bürkner, 2017).*

E.2.1

Reviewer 2 brings up the absence of a control condition, which would better support the claim you are making. I want to highlight this, as a possible way you can expand on your *study by collecting additional data. If you want to collect such data, I would welcome it, and I suspect that would be a stronger way to revise. However, I will also accept changes in your argument structure and interpretation that acknowledge and address this concern in your revision.*

Response:
Due to changes in affiliations and the expiration of the project funding, we cannot collect additional data. Reviewer 2 is specifically asking for an additional study in which the same stimuli (or part of which) may be presented to children either within the framing of a symbolic object-choice task we employed ("The monkey will help you find the banana!") and contrastively in the manner of a standard match-to-sample task ("Look at these! Which ones go together?"). We have amended the manuscript by highlighting the closeness to match-to-sample tasks in the literature and again in the discussion as an avenue for future research in addition to a paragraph from the first draft already discussing the relations between our task and analogical reasoning.

E.3. General Requirements
Together with this cover letter providing a point-by-point reply and documentation of changes made for the revision, we submit

- a clean APA style manuscript
- a track changes version of the main document
- a lay summary (page 1 of the manuscript before the abstract)

SRCD sociocultural policies
The manuscript is compliant with the SRCD sociocultural policies. Under the *General Methods* section (p.10) we report (1) the dates of data collection; (2) race/ethnicity, socioeconomic status, language, (3) place(s) from which that sample was drawn, including country, region, and report the (4) selection and recruitment procedures.

StatCheck
Both manuscript and supplement have been run through StatCheck. The main manuscript does not contain significance tests. All exploratory t-tests reported in the supplements have been evaluated as "consistent" by StatCheck.

Exploratory vs. Confirmatory Declaration
In the General Methods section under Data Handling and Analyses (p.14ff) we outline that all analyses and hypotheses were preregistered. We highlight where our analyses deviate from the preregistered analysis (i.e. comparing models via ELPD differences). Exploratory analyses are clearly discussed as such and are presented only in the supplements. We highlight that we had hypotheses about the specific order of the age of success for the experimental conditions but did not make point prediction about the month of success.

Open Science Declaration.
All data, analysis scripts and materials are publicly available. This open science declaration will be included on the title page:

> *Scientific Integrity and Openness*: The data and code necessary to reproduce the analyses presented here are publicly accessible, as are the materials necessary to attempt to replicate the findings. Analyses were pre-registered. For data, analyses materials, and preregistrations see: [LINK MASKED FOR REVIEW]

Reviewer 1

Overall evaluation

The manuscript presents 3 studies that explore children's graphic symbol understanding and use in the age range of 3-7 years. It presents a novel approach with stimuli that are more abstract than previously used in this area of research. The use of more abstract symbol-referent relations heightens the works' relevance to analogical reasoning, a core cognitive component that is very likely a core component of pictorial symbol processing. This link has not been explored in the graphic symbol literature previously. I find this work to be of high importance; it addresses an important question about the process of graphic symbol understanding and use (the role of analogical symbol/referent relations), the stimuli and tasks are carefully designed to test the hypotheses, the statistical analyses are appropriate, and the interpretations are consistent with the results. Specific points for the authors to consider are found below.

R.1.1 Consistent use of the term referent

The term referent is sometimes used to mean the symbol. This may stem from the confusion that the symbol and referent in the studies are both pictures. If this seems confusing to the authors then perhaps consistently using the terms cue and target when referring to the stimuli used in these studies, and symbol and referent when referring to other studies investigating graphic symbols, would clear the confusion.

Response:
We have amended the manuscript in line with the suggestion and now use *symbol* and *referent* when discussing the literature, but *cue*, *target* and *distractor* when discussing our work. Thanks!

R.1.2 Clarify Conceptual Framework

The literature review covers a lot of terrain, however, the links between the points is not obvious, even in the summary on p. 9. Perhaps the authors could set up the overall conceptual framework that motivated the study at the beginning of the introduction, followed by the general aims, (what is novel and why is it an important question) then delve into the specific aspects that are as yet understudied in the context of graphic symbol development. In *my reading of the manuscript the unique contribution of the work is its relevance to the link between symbolic development and the cognitive skill of analogical reasoning, but this is not sufficiently developed in the introduction for that link to be obvious.*

Response:
A systematic investigation of a comprehensive set of symbol-object-relationships tracking development continuously across the preschool years while using one coherent paradigm across several studies is our contribution to the literature. We are thankful for the suggestion to emphasize the link between symbolic development and analogical reasoning particularly. The introduction has been reworked and substantially shortened and instead explicitly treats the literature on relational and non-relational match-to-sample tasks as a point of reference for our contribution (see pages 8-9).

R.1.3 Cohesion of Introduction

In general, the flow of the points in the introduction needs to be improved. For example, on p. 4 the paragraph shifts from a review of studies where pictures are used to find objects (Callaghan, DeLoache work) to the point that pictures as symbols of pictures are rarely found. It is not clear why this is an interesting question.

Response:
Thank you, we have rewritten the introduction to improve its flow. This shift in topics has been deleted. After an overview reviewing the onset of symbolic competencies up to the third year, we first introduce the concepts and cognitive inferences relevant symbolic object-choice tasks. The literature on specific symbol-referent relationships is not reviewed stepwise but once in the context of symbolic communication and once in the context of match-to-sample designs.

R.1.4 Olson 1994

Olson (1994), The world on paper, is a seminal work on literacy and cognition which may be helpful for the authors to consider.

Response:
Thank you, this is very helpful. A central claim of Olson is that literacy provides access to new concepts and modes of thinking, with a strong focus on benefits for children's developing (meta-)analytic reasoning and linguistic awareness. We cite Olson (1994) now in the opening paragraph when referring to language and literacy acquisition as significant achievements in cultural learning (see page 3).

Olson, D. R. (1994). *The world on paper: The conceptual and cognitive implications of writing and reading*. Cambridge University Press.

R.1.5

Using the term 'realistic representation' is confusing – 'replica object' is typically used to denote these referents.

Response:

We replaced the phrase "realistic representation" in the discussion section with "*replica objects*". The introduction has been shortened considerably and the reference can now be found on page 5.

R.1.6

There are several points where clarification is needed about why a given statement is true. E.g., why are the perceptual dimensions of color, quantity, size etc easier to process than relational properties or orientation, position? Is this a perceptual mechanism at work? Or attentional? Is this mechanism operating at this age range?

Response:

Following the request to shorten the introduction by 500-1000 words, we have now omitted the statement entirely.

R.1.7

Delving more into the roots for your hypotheses – they do seem reasonable from classic perception and attention developmental work – is needed for the roots of the rationale to become clear. Some articles that may be informative along this line of reasoning are works that showed a shift from 3 to 5 to 7 years range in children's attention to whole objects vs. features of objects https://doi.org/10.1016/0022-0965(87)90057-9 and research that showed interference in the processing of hue and form features early in the attentional process (pre-attention) https://doi.org/10.1016/S0166-4115(08)60456-2 . While not directly relevant as the current task is very different, these classic works may raise additional interpretations of the object/feature findings reported here. My understanding of the early work, in a nutshell, is that younger children process stimuli as wholes and only later are their attentional processes developed sufficiently to focus attention on specific features. This developmental trend, and interference when the task requires selective attention on a specific perceptual dimension of a stimulus (size, orientation, form, etc) may explain, for example, why the size of the feature was particularly difficult for children.

Response:

Thank you, this is extremely helpful. We have incorporated these references in the interim discussions concluding the individual studies and refer to the shift from holistic to feature-oriented processing also in the discussion (pages 22, 26, 30, 34).

Callaghan, T. C. (1990). Texture segregation in young children. In *Advances in psychology* (Vol. 69, pp. 175-195). North-Holland.

Shepp, B. E., Barrett, S. E., & Kolbet, L. L. (1987). The development of selective attention: Holistic perception versus resource allocation. *Journal of Experimental Child Psychology*, *43*(2), 159-180.

Diamond, A., Carlson, S. M., & Beck, D. M. (2005). Preschool children's performance in task switching on the dimensional change card sort task: Separating the dimensions aids the ability to switch. *Developmental Neuropsychology*, *28*(2), 689–729. https://doi.org/10.1207/s15326942dn2802_7

R.1.8

The delineation of hypotheses was helpful, but it would also help to understand why these predictions were made. Reworking the introduction may make the rationale for the hypotheses more transparent.

Response:

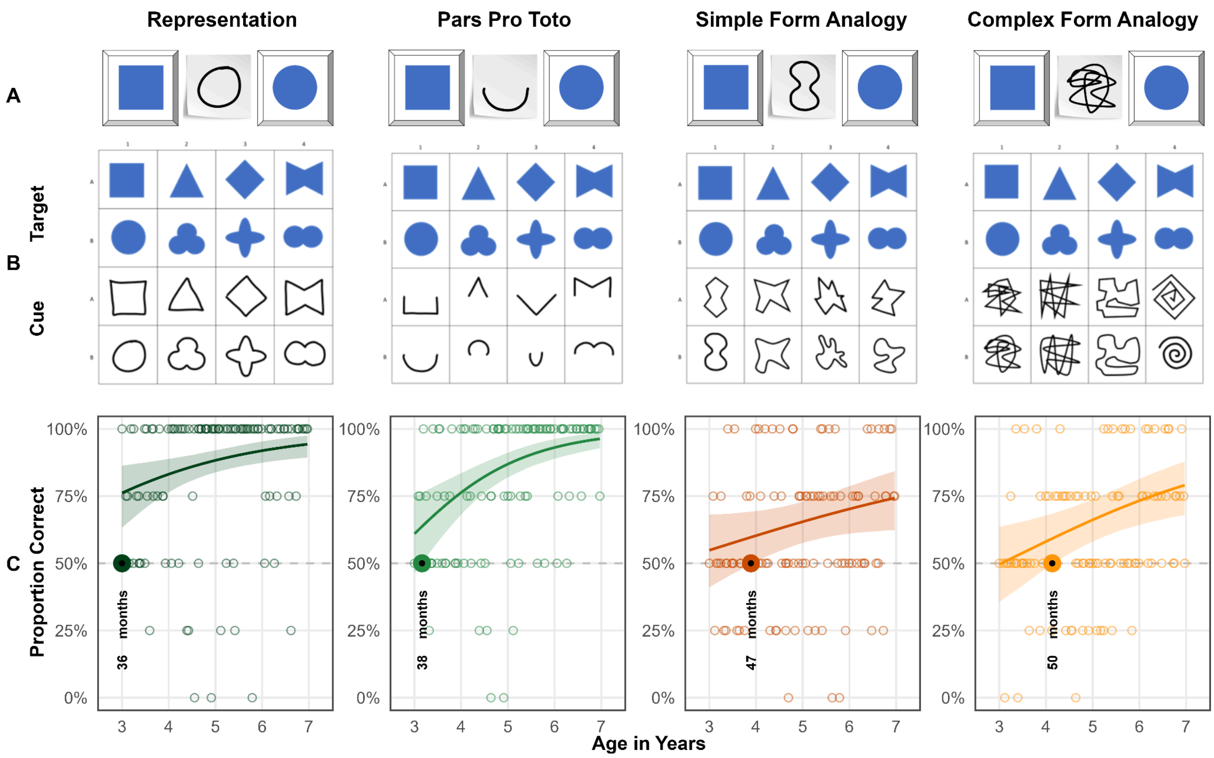
Thanks for pushing us to be more explicit here. We hope that the reworked introduction makes the motivation and basis for the hypotheses more clear. Our main aim, however, was to contribute to the literature by testing various symbol-referent relationships in one coherent paradigm and trace development through the preschool years. To be more explicit regarding the preregistered hypotheses but without adding to the word count of the introduction, we now provide short justifications of the hypotheses when stating them at the beginning of each study section on pages 17, 22 and 26ff..

R.1.9

For all figures it would be helpful to have the h/v axes labelled for B and C panels.

Response:

We have amended the illustrations accordingly, e.g. for study 1:



R.1.10

p. 6 – I did not understand the point in the last line of the paragraph beginning 'A highly abstract…'.

Response:

This is from a paragraph introducing analogies in form (round/edgy). The sentence has been removed and the introduction was reworked. Also and in line with suggestions from reviewer 3, the discussion of sound-symbolism has been shortened considerably. The paragraph now reads:

> *Another way of reducing iconicity may be the metaphorical use of form, such as using sharp or spiky drawings to refer to "pointy" concepts (Knoeferle, Li, Maggioni, & Spence, 2017). This phenomenon is central to research on sound symbolism within the lexicon (Bermúdez, 2020; Sidhu, 2025), as well as in studies of action and gesture (Bohn, Call, et al., 2019; Margiotoudi & Pulvermüller, 2020). Robust across cultures (Ćwiek et al., 2022), this sensitivity emerges around 30 months for word-shape matching (Fort et al., 2018; Imai, Kita, Nagumo, & Okada, 2008; Maurer, Pathman,*

& Mondloch, 2006) and by 36 months for matching actions with gestures (Bohn, Call, et al., 2019). Despite these findings, it remains unknown whether children can utilize the global properties or "impressions" of a shape's form to ground reference in the graphic domain alone.

R.1.11 *p. 7 – 'size words' is repeated*
Response: amended

R.1.12 *p. 9 – 48-60 months could be 4-5 years as years is the age unit throughout the paper.*
Response:
Thanks! We now use years wherever possible, except for when discussing changes occurring between 30 and 36 months when referring to the work by Judy Deloache and Tara Callagher, or when presenting our own findings.

R.1.13
p. 10 – it is misleading to indicate there are 96 children in the general methods as different numbers of children are tested in each of the studies. An important point of clarification is whether different children were tested in each study.

Response:
Thanks! During data collection, we have ensured that children can only participate once in any study contributing to this research project. Hence, different children were tested in each study. We had preregistered to collect a minimum of 96 children per study. We have amended the manuscript to make this more clear, for example, in the abstract:

> *Participants were <u>304</u> German-speaking children continuously sampled from age 3 to 7 (Study 1, N = 106, 51 female; Study 2, N = 99, 49 female; Study 3, N = 99, 55 female).*

as well as the general methods section:

> *In order to continuously trace the development of symbolic competences across the preschool years, for each study we aimed to test <u>a minimum</u> of two children per month of age between the third and the seventh birthday for a total of 96 participants balancing male and female participants.*

and when discussing the sample of study 2 and 3:

> *A total of 99 three- to seven-year-old children (M = 60.04 months, SD = 13.69 months, range 36 - 83 months; 49 female) participated [in study 2]. <u>None of these had participated in study 1.</u>*

> *A total of 99 three- to seven-year-old children (M = 59.88 months, SD = 13.44 months, range 36 - 83 months; 55 female) participated [in study 3]. <u>None of these had participated in the previous studies.</u>*

R.1.14
p. 14 – it is on this page that the use of referent became confusing.

Response:
Again, thanks for pointing out this lack of consistency. We now use *cue*, *target* and *distractor* throughout the manuscript.

R.1.15 *p. 15 – children's choices were logged by experimental script, or experimenter?*

Response: Children made their choice by touching either hiding place on screen. The experimental script immediately recorded which side they chose and choices were immediately logged as correct or incorrect according to the logic of the task. The manuscript was amended here to state this more explicitly:

> *Upon touching the screen, children's choices were logged and immediately coded as correct or incorrect by the code running the stimulus presentation.*

R.1.16
p. 18 – I wondered about the ages of children excluded – and see in the supplementary the breakdown of this. For Figure S1 – are excluded children included or excluded from this figure?

Response:
There is no overlap between the children in supplement figure 1 showing the ages of participants and supplement figure 2 showing the distribution of excluded children across the age-range. We have amended the manuscript by referring to the two supplement tables as indicating children included and excluded respectively.

R.1.17
Reference is made to the 'reference condition' throughout the results sections – it would help to specify the comparison condition as there are many conditions to track throughout the paper.

Response:

Yes, thanks! The general methods section states that the hypothetically most simple condition in each study was used as the reference condition in each model (study 1: representation, study 2: absolute position, study 3: size of object). When discussing the specific results we now suppress the ambiguous term "reference condition" but name the respective conditions explicitly.

R.1.18

p. 21 – In reference to the Simple and Complex Form Analogy stimuli, saying they are based on visual similarity seems to be a stretch. For example, sharing curved lines may link the drawing of a flower and a VW Beetle, but they would be judged as visually dissimilar.

Response: This is a very interesting point. Our experience from presenting this work in professional contexts is that researchers have very different intuitions about what iconicity or visual similarity are. While - as expressed here - some may be hesitant to refer to an "analogy in form" as visual similarity, others have discussed all stimuli we used as "being merely iconic", even including size and number .

To amend the paragraph we avoid this phrasing now and have replaced the sentence with:

> *While they are still based in iconicity, in so far as their interpretation involves a sense of resemblance with regard to their shape (Winter, Woodin & Perlmann, 2026), they are conceptually demanding at the same time.*

This is drawing on a recent contribution to the Oxford Handbook of Iconicity by Winter and colleagues (2026) suggesting a highly inclusive definition for iconicity in the cognitive sciences like "*A signal in any modality or medium exhibits iconicity when its production and/or interpretation involves a sense of resemblance between at least some aspect of its form and at least some aspect of its meaning.*" Given the more round or edgy shapes used in the form analogy conditions, there is at least a sense of resemblance to some aspects. Of course, there is a continuum from highly iconic to more conceptual symbol-referent relationships that we wanted to refer to when discussing the results.

R.1.19

p. 24 – My sense is that the condition Orientation of Feature would be better labelled Position of Feature as it has a top/bottom options and not directionality.

Response:

We have specifically opted for using only top/bottom options here as data from another study that we are currently preparing for publication is suggesting that children do indeed read such displays with a salient feature (a shape with a bump, dent, opening on one side) as being directional. Against this background, we would like to take the liberty of not changing the manuscript here. In advance, thanks for your understanding.

R.1.20

p. 25 – 'assign or abstract' – would 'infer' be a more apt term?

Response: The phrase has been replaced in line with the suggestion of the reviewer:

> *In order to correctly interpret the symbol-referent relationships employed here, children need to infer conceptual order in graphical displays.*

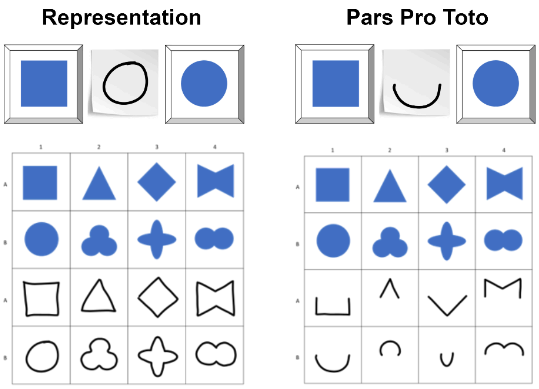
R.1.22

p. 31 – I do not understand the phrasing 'While cues are highly similar and targets even identical..'

Response:

Sorry for the confusion! The sentence compares the stimuli used in *Representation* and *Pars Pro Toto*. The targets (upper two rows in blue) are identical in both conditions and the cues (bottom two rows, outline drawings) are cut in half. The amended paragraph now reads:

> A second condition, *Pars Pro Toto*, used the same stimuli as *Representation*, but the cues provided only half of the respective figures to guide children's choices (cf. Figure 2 B, boxes 1 & 2). Despite using the same target stimuli as *Representation* and very similar cues, children succeeded in *Pars Pro Toto* no sooner than at 38 months of age.



R.1.23

p. 33 – the link between the trajectory of EF skills development and the current findings needs to be fleshed out more, It is difficult to see the link as it is phrased. In general ,these points are very interesting interpretations – especially reference to analogical reasoning work, but without more framing in the Introduction readers will need to take these statements at face value.

Response:

We have now added references with examples for how EF skills and communicative and analogical reasoning interact. We agree that being more precise here adds to the manuscript but would like to keep it as brief as possible to comply with requests for shortening the manuscript, but also since the importance of EF skills in match-to-sample, reading, drawing and decoding graphic displays is well documented.

In the introduction we highlight the role of EF skills in match-to-sample tasks and added references (see pages 8-9). In the discussion, we outline how components of EF may help with the demands of our task:

> *Specifically helpful for the tasks employed here may be developmental gains typically occurring around the fourth birthday in the domain of analogical reasoning (Christie & Gentner, 2014; Gentner, 1977; Shivaram et al., 2023), which are again fostered by a suite of executive function skills improving significantly during this important period in child development (Bohn, Tessler, Kordt, Hausmann, & Frank, 2023; Thibaut, French, & Vezneva, 2010). Focusing on various aspects of the stimulus-combinations, working memory (Richland et al., 2006; Simms, Frausel, & Richland, 2018) is crucial for keeping track of features children have explored. To succeed they are furthermore required to suppress irrelevant aspects of the stimuli and inhibit distracting impulses solving relatively abstract tasks (Carlson & Wang, 2007; Jablonski, 2014). Finally, cognitive flexibility is not only key for viewing the same stimulus from different conceptual angles but also specifically for switching modes of interpretation across conditions (Cuevas et al., 2014; Willoughby, Wirth, & Blair, 2012).*

R.1.24

p. 34 – Do the authors think that when you use highly abstract pictorial symbols in these tasks that the more abstract relations will pop out more easily than if realistic (or highly familiar shapes) pictorial symbols are used? It may be that we use basic filters as a first line of processing with pictorial symbols (color, form) because it works for most symbolic matching tasks, but pull out other filters (orientation, number, size) when the match uses more abstract symbols. The first paragraph on this page needs to be fleshed out – very interesting suggestions that raise more interesting possibilities.

Response:

Our hunch is that using realistic target stimuli of familiar objects will make it harder for children to find abstract relations. One reason for this may be that depictions of natural objects are necessarily more complex than simple geometric shapes. In addition, natural objects will trigger semantic processes (context, affordances, use cases). In both cases, cognitive load increases and may tax children's performance. An interesting case in question that we are citing is Long et al.'s (2019) stroop-effects when asking preschoolers to make size comparisons with depictions of real-world objects that are congruent or incongruent with their actual size. The paragraph in question has been amended using the references suggested above (Shepp, Barrett & Kolbet, 1987; Diamond, Carlson & Beck, 2005).

> *While Size of Object demonstrates when children can grasp size analogies earlier, Size of Feature stands out as the most difficult condition in this series of studies. Success here requires shifting attention from general appearance to nested object features and this ability develops considerably throughout the preschool years (T. C. Callaghan, 1990; Diamond, Carlson, & Beck, 2016; Shepp, Barrett, & Kolbet, 1987).*

R.1.25
p. 35 – first complete sentence is hard to understand. The working memory points seem unrelated.

Response:
We wanted to point out that searching for relations in the more abstract or complex stimuli requires children to successively focus on different aspects of target and cue (attentional shift) and keep track of the dimensions they have explored (i.e. check for congruence in form, number, orientation). This is where working memory will be beneficial and it is another aspect with crucial developments around age four. We amended the paragraph:

> Second, the reduced accuracy in analogy-based and feature-referenced tasks are both consistent with evidence that younger children tend to focus on surface features rather than relations (Gentner & Hoyos, 2017; Rattermann & Gentner, 1998; Richland et al., 2006), making tasks that require attentional shifting or nested comparison structures particularly demanding (Halford, Wilson, & Phillips, 1998). Shifting attention to different aspects of stimuli sets requires children to keep track of features they have explored. Relational or abstract as opposed to feature-based processing taxes working memory and inhibitory control more strongly (Gick & Holyoak, 1980; Goswami, 1991). This, in turn, points to aspects of cognitive maturation that are likely to foster children's performance in the tasks outlined above, as seen in work on the development of executive functions — particularly the early emergence and developmental trajectories of working memory, inhibitory control, and cognitive flexibility in childhood (De Luca & Leventer, 2010; Diamond, 2013; Reilly, Downer, & Grimm, 2022).

R.1.26
p. 36 – There is some early work on children's production of pictorial symbols showing that children use a wholistic approach and attend to features that will distinguish the referent only later, or if given a prompt that their drawing is difficult to interpret https://doi.org/10.1111/1467-8624.00096

Response:
The publication by Callaghan (1999) is relevant and we are already citing it in the introduction and discussion. In line with suggestions from the other reviewers we would reduce the amount of space dedicated to the production of graphic symbols, but highlight changes from holistic to feature-oriented processing.

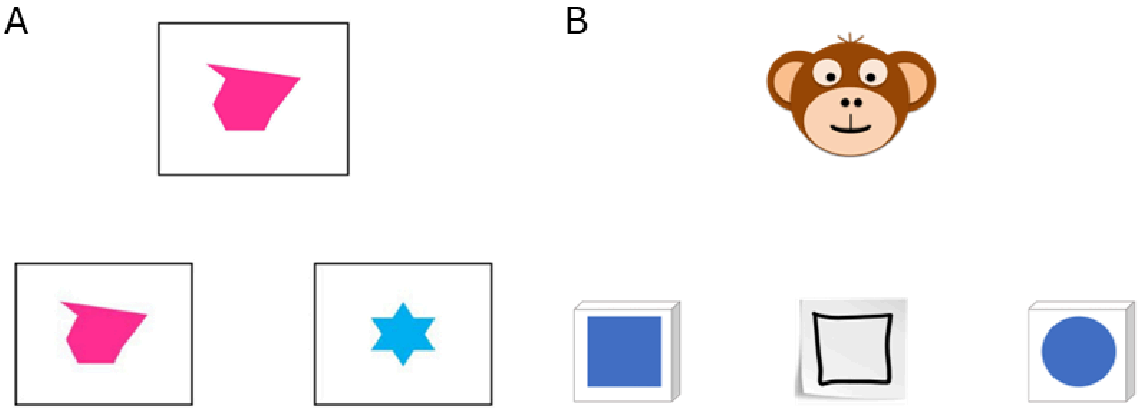
Reviewer 2

Overall evaluation
I commend the authors for the technically well-executed experiments, for the wide variety of relations tested, and for the sound Bayesian analysis methodology. My only problem with the manuscript in its current form is the untested assumption that children interpret the cue–target pairs as symbol–referent pairs. The authors require this assumption in order to argue that the results bear on children's departure from iconicity with age. I elaborate on this below.

R.2.1
*The manuscript claims to be testing "children's ability to interpret graphical symbols" (p. 2). On its own, this sentence is ambiguous between a de dicto reading (children's ability to interpret these visual stimuli, which happen to be graphical symbols) and a de re reading (children's ability to interpret these stimuli *as symbols*). Clearly, they have in mind the second reading: "Our work provides one of the most comprehensive, robust and systematic investigations of young children's ability to find meaning in unfamiliar graphic symbols" (p. 36).*
I suppose the authors assume this interpretation to be likely because of the way the task is set up: a cooperative agent creating drawings on the spot for informative purposes. But there's nothing in the data that supports this interpretation. Indeed, the authors' experiments can be redescribed as match-to-sample tasks just as well. In the absence of a control condition with different results (for instance, one in which there are no drawings involved), there's no way to know whether it matters that children interpret the cues as symbols, or whether they just select the target that exhibits the higher similarity (where similarity is computed along many dimensions) to the cue. In this case, what develops with age is sophistication in how similarity functions are computed. This development might be relevant to how children interpret symbols outside the lab, but it would be a domain-general capacity not specific to symbolic communication.

Response:
We are very thankful for the opportunity to discuss this concern as part of our revision and have rewritten the introduction and presentation of the task to address the problem at hand. We now provide a step by step overview of what is required of children to solve the task in our studies (see page 4).

In a standard **match-to-sample task**, children are presented with two choice items, *match* and *distractor*, and invited to take a closer look. Next, the *sample* item is placed in the middle with the prompt: "Which one of these two goes together with…*experimenter placing the sample* …this card?" (cf. Kroupin & Carey, 2022). Here, the children are directly instructed to identify a match and this is the only inference required. In a **symbolic object-choice-task** children are presented with a hiding game in which a reward is hidden under one of two containers, one serving as the target and one as a distractor. Next, the experimenter directs the child's attention to the task ("And now…look!") and provides a communicative cue. Then the children are provided access to the hiding places and prompted to make a choice ("Do you know where it is?").



A: Typical Identity match-to-sample trial as used in Kroupin & Carey (2022); B: Illustration of a choice trial in symbolic-object-choice task.

In the standard analysis of this setup, children need to understand (1) the pragmatics of the hiding game, (2) that there is a cooperative partner in front of them intending to help, and (3) that the partner is doing so by means of a communicative cue. As a final step, children need to comprehend the cue and connect it to either choice option. In studies on communication, communicative cues may be pointing, markers, replicas, drawings, iconic gesture or switching on a light (Callaghan, 2000; Tomasello, Call, Gluckmann, 1997; Moore, Liebal & Tomasello, 2013; Bohn, Call & Tomasello, 2016). All of these cases may be criticized as merely capitalizing on spatio-temporal contiguity (deixis) or (crossmodal) matching, but they still require communicative inference and pragmatic reasoning.

We are confident that our work provides a crucial contribution to the literature. It exceeds previous work on graphic communication by systematically testing various symbol-referent relationships with robust samples and it goes beyond match-to-sample by testing MTS relations in a communicative context. Please consider also that conditions such as Pars Pro Toto, and the Simple and Complex Form Analogy have no equivalent in the MTS literature. Also we use outline drawings as cues that are more abstract than presenting the cut-out stimulus twice. To cater to the points of the reviewer, we are now devoting more space to discussing the closeness to match-to-sample designs in the introduction by comparing the paradigms (see page 9) at the expense of shortening the discussion of work on graphical communication. Also, please consider this paragraph from the discussion section already present in the first submission:

> *While work on analogical reasoning abilities has highlighted that preschoolers are generally able to identify agreement in graphic displays and specifically involving size, number, and shape also at the age of four (Kroupin & Carey, 2022b, 2022a), this has mostly been established in match-to-sample paradigms where children are conventionally presented with three items and prompted with a phrase such as "which ones go together" directly inducing the cognitive operation tested. In our task, children make matches and comparisons across various dimensions without any verbal scaffolding addressing the match-making process or the dimension of comparison and, hence, showcases abilities for implicit judgements across a wide variety of basic conceptual and visual features. Therefore, our work provides a crucial extension to the literature on analogical reasoning by showing that children can actually put their developing skills to work in the practical context of a versatile communicative coordination game requiring both relational reasoning and pragmatic inference.*

R.2.2

Do the authors expect different results if children were given the same stimuli and asked instead to select the one that is more similar? If yes, then the authors have to argue that there is no other way to account for the results than by positing that the implicit nature of the task reveals interpretation of the cue as a symbol and of the banana container as its referent. If not, what would be the different results and why? [Counterarguing that there is extensive evidence that children interpret drawings as symbols would only go so far, since there are no actual drawings in the study—the cues are, in fact, representations of drawings, which children must infer from the schematic animations on the tablet: by this age, they know very well that there is no actual drawing monkey in front of them.] Let me state explicitly that this does not draw anything away from the findings themselves, which are interesting regardless of how they are framed. But, unless I am missing something, it unclear to what extent the findings can inform symbol–referent relations in cognition.

Response:

We appreciate that the reviewer acknowledges that this is an issue regarding the framing of this work and that it does not draw anything away from the findings. As we are now explicitly outlining in the abstract and introduction, our task requires both analogical reasoning (match-to-sample) and pragmatic/communicative inferences. As such our work connects both strands of literature.

Much like reviewer 2, we were aware of the tension and very interested in addressing this intersection directly. In an ongoing project, we are currently comparing the influence of communicative framing (symbolic object-choice-task) and non-communicative framing (standard match-to-sample) on children's performance with graphic stimuli. Preliminary results suggest that preschoolers perform considerably better when the same classic match-to-sample stimuli are used in standard framing as opposed to a communicative framing. Our preferred interpretation is that the communicative framing is more demanding as it requires pragmatic inference (keeping track of the context, not being told what to do). However, as of now these results are still pending.

In addition to having reworked how the studies are presented in the introduction, we now also address this in the discussion as a goal for future investigations.

> *The research described here has limitations that future research needs to address. The combinations of outline drawings and geometric shapes we used here could also be presented in a standard match-to-sample procedure in which children are solely asked to find a match between cue and target. When directly instructed to align stimuli and without the need for considering context, children may succeed earlier. To connect the work presented here with the literature on analogical reasoning, future work should present children with the same set of stimuli in the context of both paradigms.*

R.2.3
The authors make an observation in passing that is relevant for this question: "Success even prior to the fourth birthday in these conditions is remarkable as it requires children to identify and compare relations (Absolute Position – figure/ground; Relative Position – figure/figure) which is demanding even later throughout the preschool years in other contexts (Shivaram, Shao, Simms, Hespos, & Gentner, 2023)" (p. 32). If struggle with relational matching children in nonsymbolic contexts but find it easy in symbolic contexts (assuming identical tasks), the authors' argument would be much strengthened. However, the CogSci proceedings paper the authors cite reports successes, not struggles, on relational match-to-sample tasks in 4½-year-olds.

Response:
Yes, Shivaram and colleagues (2023) report success in 4½-year-olds, but specifically against the background of failure of the same age-group with highly similar stimuli in Christie and Gentner (2014) and Kroupin and Carey (2022). We add all three references but keep the phrasing because RMTS is indeed demanding for preschoolers and results are mixed. As for the comparison of our findings and the RMTS literature, the stimuli in *Absolute Position* and *Relative Position* are relational by drawing on Gestalt principles (figure-ground; proximity) but are still more simple than classic RMTS tasks or the stimuli in study 3. This is what we wanted to draw attention to as this is again a valuable contribution for researchers interested in graphic communication, map-reading, Gestalt-Principles but also RMTS.

R.2.4

I can think of several ways to address this: 1) If I missed something, point it out (and modify the manuscript such that other readers don't fall into the same trap). 2) Find evidence in the current data or arguments from the literature that children do interpret the visual cues as symbols (e.g., because they perform better on these tasks than on more standard match-to-sample ones). 3) Adopt a more moderate position, and claim that the experiments test relations of various sorts systematically, then spell out the potential implications of the findings for symbolic competence, iconicity development, and so on in the General Discussion. This option would require de-emphasizing the literature review on symbols in the Introduction.

Response:

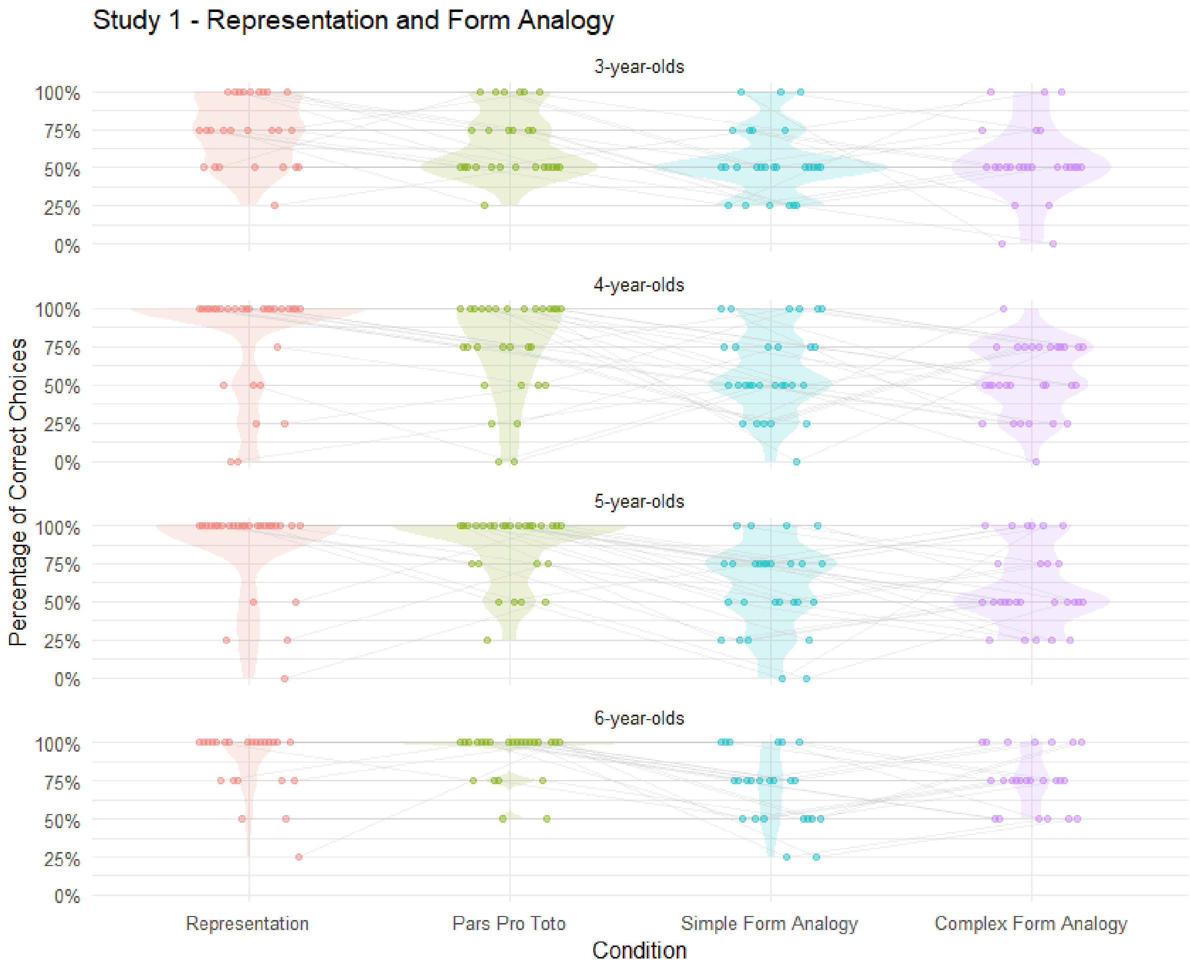
Any object or action can be a symbol or utterance when used with communicative intent. In the context of the symbolic object-choice-task we use, it would be misguiding to not use the term symbol-referent-relationships when discussing the many ways that motivate a child to see how cue and target are matching, and there is hardly a trap here: the introduction - in its original form and more so now - specifically refers to match-to-sample tasks several times, we frequently highlight that the relationships we test are based in analogies and the discussion also devotes several paragraphs referring to analogical reasoning and match-to-sample tasks, including this paragraph explicitly addressing the point under discussion.

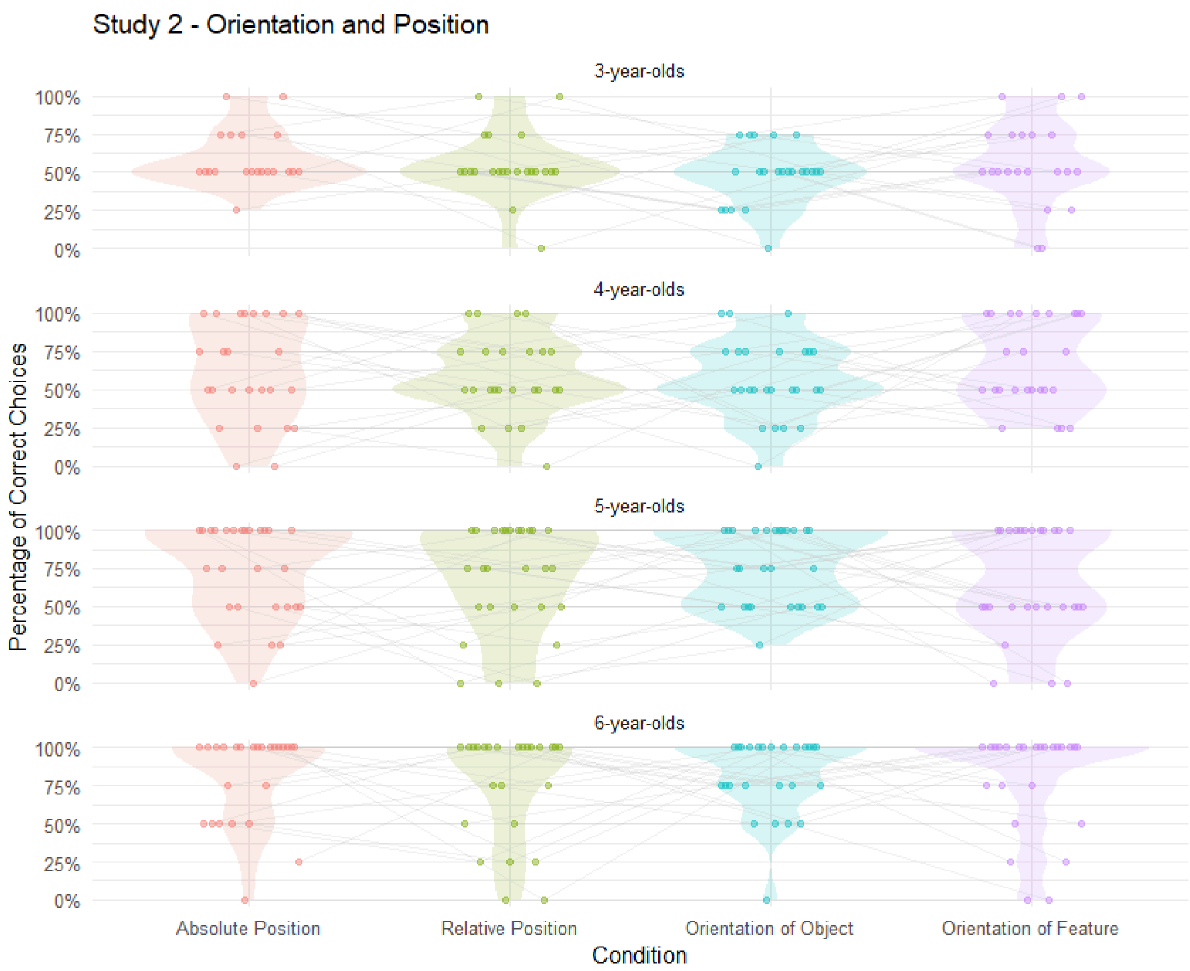
R.2.5 Data visualization

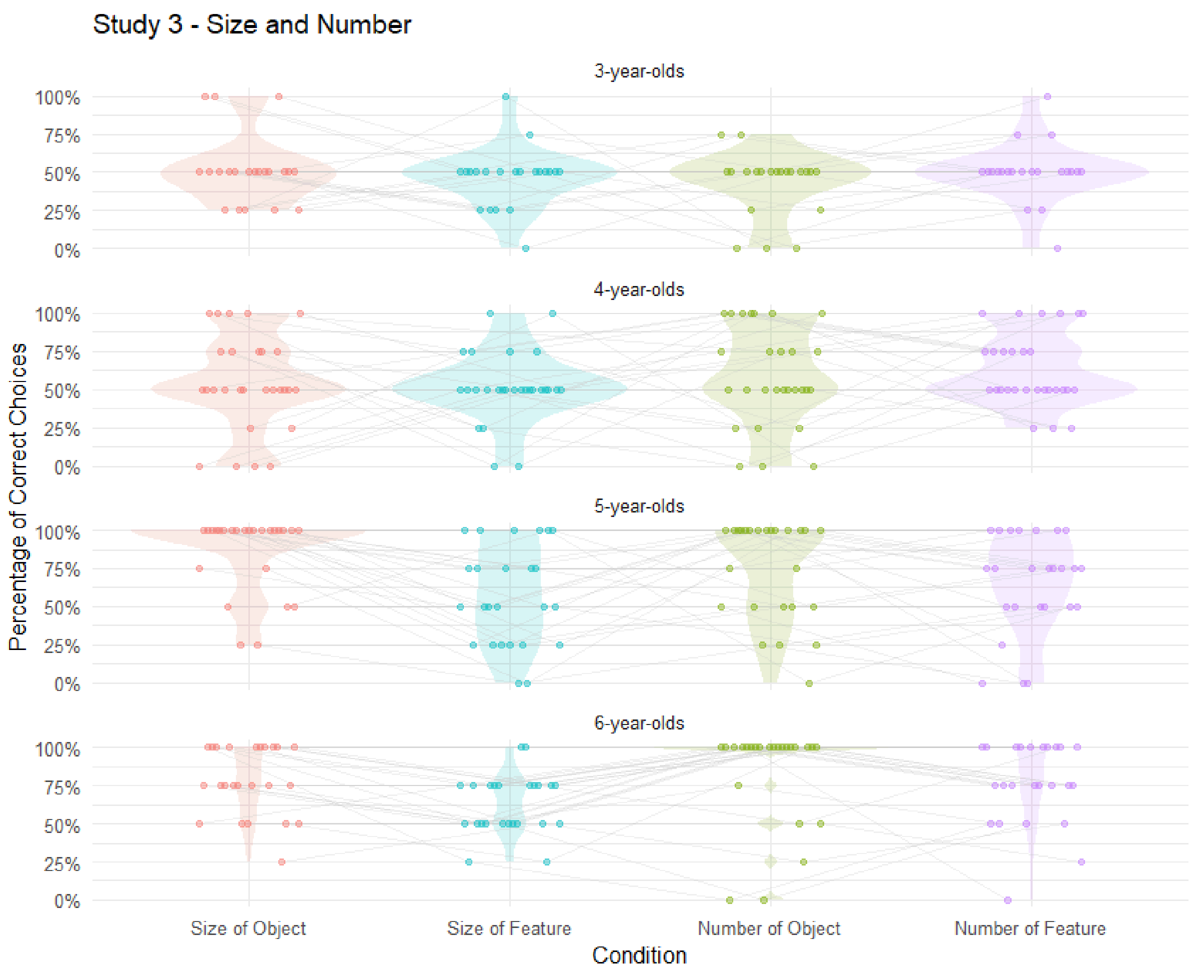
The results figures for all three experiments are currently plotted between-subjects, as it were (since they plot the results in each condition separately). But since condition was within-subjects, it would have been great to have plots illustrating performance in each condition by age group (one plot/facet for 3-year-olds, one plot/facet for 4-year-olds, etc.). Each of these subplots would show the performance of individual participants across conditions (from the most iconic to the most abstract), connected with lines. This would show at a glance how the iconic–abstract gradient affects performance within single subjects. Whether supplementary analyses are required is hard to tell (perhaps the data is clear enough that visual inspection alone will do). I am sure the authors can make a sensible decision about this.

Response:

Below, please find plots faceted by year of age. In the paper we avoid binning participants by year of age as this would entail losing information. Our goal was specifically to sample participants continuously throughout the preschool years and draw smooth developmental curves for the respective conditions. We then decided to facet by condition (cf. figure 2,3,4) as there was too much overlap between conditions. To us, no clear picture emerges from the three novel plots below, which is why we would opt against including them in the manuscript or supplements.




Study 2 - Orientation and Position
3-year-olds
4-year-olds
5-year-olds
6-year-olds
100%
75%
50%
25%
0%
Percentage of Correct Choices
Absolute Position
Relative Position
Orientation of Object
Orientation of Feature
Condition



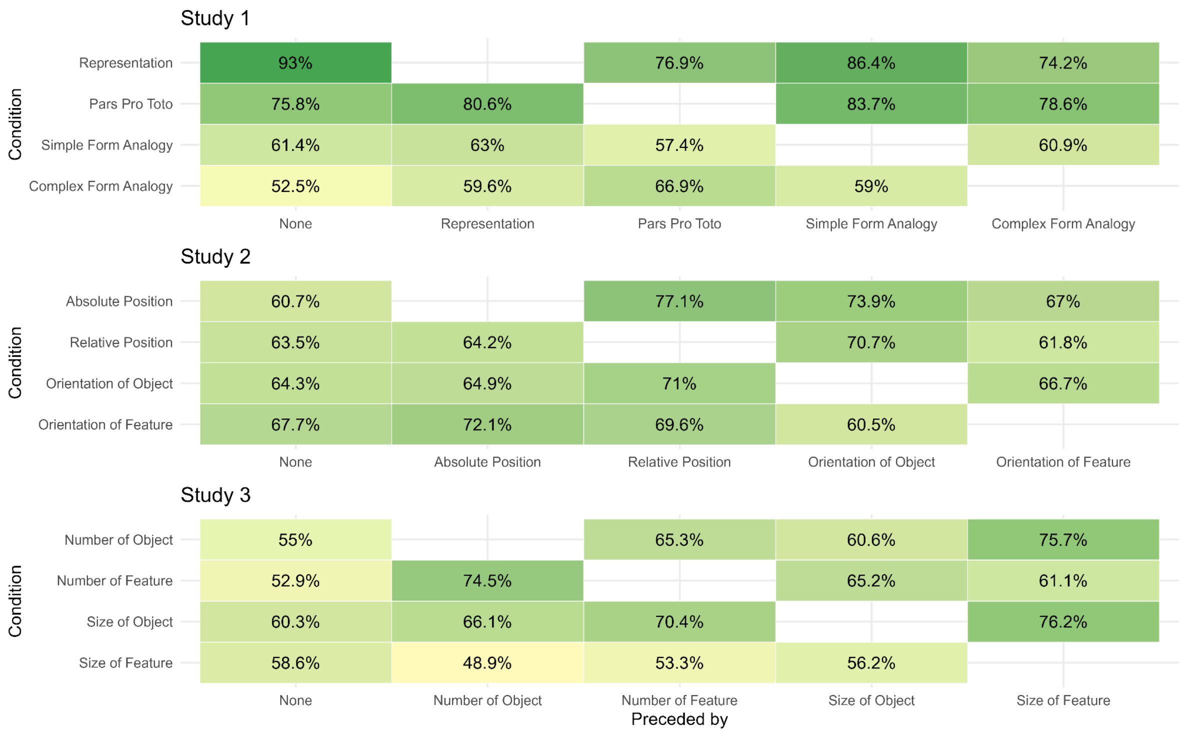
R.2.6 Supplementary analysis

The task of Study 1 has one pragmatically curious aspect, which may affect performance in interesting ways. Here's what I mean: In Study 1, the targets are repeated across conditions (e.g., circle versus square), which makes the monkey's decision to opt for different drawings (despite communicating the same content) somewhat strange. Could this contribute to younger children's struggle with the task? One potential way to address this would be to check whether there are order effects and whether they interact with children's performance from earlier to later trials: For instance, are the children who saw the representation condition first more likely to err on pars pro toto, simple-form analogy, and complex-form analogy later on? (To illustrate: If the monkey is clearly able to draw a circle, why does she turn to a scribble? This could give rise to a mutual exclusivity inference and thus to selecting the 'wrong' answer.)

Response:

To an extent, the reviewer's comment applies to the experimental situation in a large amount of studies on communicative development: Isn't it odd to solely point silently instead of telling the child where the toy is hidden? Why is the experimenter using iconic gestures instead of telling the child how to solve the riddle box? In our case, the reviewer highlights that the repeated use of similar stimuli may be particularly problematic. Generally, as there is always the possibility to provide a clean drawing of the correct target, the monkey is violating the Gricean Maxime of quality (be as informative as possible) and/or manner (be clear and orderly; avoid obscurity) in all conditions except for *representation*. We think that this is necessary for testing specific symbol-referent relationships in isolation but in a within-subjects design and not irritating in the context of a playful picture-book-style developmental study.

To address order effects, we have aggregated data from all studies and provide the percentages of correct choices for each condition (y-axis) separately depending on the preceding condition (x-axis). For example in study 1, *Representation* has the highest performance when presented in the first block. *Pars Pro Toto* has a slightly higher proportion of correct choices when preceded by *Representation*, which is reasonable as it is just the reduced version. At the same time *Pars Pro Toto* has its highest values when preceded by *Simple Form Analogy*, for which I can provide no good explanation. *Simple Form Analogy* yields almost identical results regardless of its position in the counterbalancing and *Complex Form Analogy* is hardest in first position and benefits more from being preceded by *Pars Pro Toto* than *Representation*. So at least it is not detrimental to children's performance to go from a condition with clear "utterances" to another one violating Grice's maximes. All in all, the picture that emerges is not clear and switching effects appear moderate overall.



Please consider, the overall developmental trajectories we report as our main finding may not be influenced by switching costs as variance due to trial order has been controlled for and the order of conditions is counterbalanced. It is likely that EF skills may impact switch costs at an individual level, which is a good justification for including random slopes for individuals as we did.

R.2.7
In Study 3, there are no target repetitions but the symbols are shared between condition pairs (size of object – size of feature and number of object – number of features). Are children more likely to make a mistake when a symbol is repeated?

Response:
In both cases, children are actually performing better when the conditions are preceded by each other than when occurring in the initial position (see plot above). When doing two size or number blocks successively, this helps children as it may reduce the cost of switching across tasks. Children do not appear irritated by symbols being shared across conditions.

R.2.8
Finally, also in Study 3, the crossing of the square and the circle also becomes pragmatically odd, as children progress through the task: If the monkey can draw a circle (on trials with squares) why does she not draw one on circle trials? Evidence of pragmatic inferences could, I think, also bolster the authors' claim that children interpret the drawing as symbols of the referent locations (at the very least, it would show that children interpret the drawings communicatively).

Response:
Reviewer 2 suggests that the crossing of circles and squares: a big circle referring to a big square in contrast to a small square, and then vice versa (cf. the first two columns in the matrices below) may be particularly confusing or demanding for children in consecutive trials. We are very thankful for this thorough reading of the manuscript and for taking the time to actually picture what participating in this study may be like for children. However, we see no issues here with regard to the study methods or the interpretation of results.

First of all, this monkey is not a good Gricean communicator and this is a compromise we were willing to make when designing the study in order to test relatively abstract symbol-referent relationships. We now acknowledge this directly in the introduction when discussing the setup (page 6).

Generally, counterbalancing the order of conditions should ensure that all conditions may be equally affected by potential effects of order. We have opted to cross circles and squares across trials in study 3 as they are the most simple and canonic shapes - hopefully reducing processing demands - and for also being the most distinct: matches between cue and target may only be made on the basis of size but are completely implausible with regard to shape.

If there would be problems with regard to trial order, either trial version should be equally affected as the order of trials and conditions were counterbalanced. The problematic transition under discussion has a 1 in 4 probability of occurring. Furthermore, children would have to track stimuli across trials to be irritated by this. If so, they may for example get frustrated and stop earlier as the monkey appears to be an irrational communicator. However, drop-out rates are negligible (study 3: one exclusion for not completing at least eight out of 16 test trials; three exclusions due to fussiness) and results are controlled for trial order.

We think it likely that children actually enjoy such communicative games when containing puzzles. Much like when playing charades where not being able to talk is particularly exciting, not using the most obvious way of graphic reference in the context of our hiding game may be interesting too. Children are also very familiar with not directly being provided with solutions in hiding games, but rather receiving playfully suggestive cues.

R.2.9 A potential reason for the difficulty of some conditions

I was also wondering whether it matters that children received each target pair only once in each condition, paired with a single cue. So children lacked access to the contrast between the two alternatives the authors designed. I guess the authors expect that performance would improve if children witness the monkey choosing between the two options? (It is clear to me, for instance, which cue goes with each target in the complex-form analogy condition of Study 1 because I can clearly see the contrast between the round and the spiky cue, but I am not sure I would have converged on the 'correct' interpretation without it.) It seems that this is worth mentioning in the General Discussion.

Response:

We agree that this makes the tasks more difficult and it was an intentional choice to present children with a novel cue-target pairing in each trial (and in the absence of feedback etc.). We think this is ideal for investigating which symbol-referent-relationships children can decode spontaneously. Providing the contrasts in question would have made it significantly easier but also changed the task considerably. But consider that children saw four trials per condition in a blocked fashion. For example, in the *size condition* this would entail seeing four cases of the monkey drawing a novel cue that was either large or small to refer to targets differing solely in size. Instead of adding to the discussion that performance could have been increased by presenting pairs of cues that the monkey could choose from (or providing feedback, training match-to-sample first, providing more verbal scaffolding, etc), we would take the liberty of adding to this point to the discussion when highlighting why our paradigm is a conservative test of children's abilities (page 31).

Reviewer 3

Overall evaluation
The research questions examined are theoretically relevant and the findings will be of great interest to developmental psychologists. The presentation of the manuscript could be greatly improved to more clearly motivate the research questions and situate the findings in the literature, as I will detail below.

R.3.1
1. I understand why the authors use the term "graphical symbols" as the paradigm involves the generation of the symbols by the monkey. I think it would be helpful to more clearly outline the use of the term early in the manuscript to help the reader appreciate what is meant by graphical.

Response:
We have reworked the introduction and added a paragraph providing a detailed discussion of symbolic-object-choice tasks together with the inferences required by children to solve these tasks. Here we list various types of cues and introduce the term "graphic symbols" (see page 4).

R.3.2
2. In a number of places in the manuscript, including the keywords, the term pragmatics is used. I am not sure in what sense this term is being used in this context and it would be helpful to clarify early on in the manuscript.

Response:
Thanks! The introduction now contains the aforementioned paragraph providing a set-by-step analysis of the inferences children need to make when solving the symbolic object-choice tasks we used in our tasks. Here we specifically introduce *pragmatic inference* as a way of referring to the step where children have to integrate the demands of the search game, the perceived helpful intent of their interlocutor, and the perceived similarity between the interlocutors drawing and one of the hiding places, to arrive at the conclusion that the drawing is a cue referring to a target (see page 4). Hence, pragmatics here does not narrowly entail linguistic concepts such as implicature (Grice) or illocutionary force (Austin) which may also have been plausible associations for readers with the term *pragmatics*.

R.3.3
Much of the literature review focuses on children's interpretation of drawing, which is of course, highly relevant to the research questions. The discussion of the role of shape could, however, be more concise (for example, I am not sure why there is discussion of shape and sound-symbolism), with more space devoted to discussing the mechanism that underly children's interpretations of drawing. For example, I expected more consideration of work by

Gentner and colleagues around comparison and analogical reasoning, given the paradigm used invites children to engage in comparison reasoning to identify the correct target.

Response:
Prior to revision our goal was to provide an in-depth overview of children's production and comprehension of the symbol-referent-relationships tested in the three studies, but we agree with the editor and reviewers that this was not ideal. We have restructured and shortened the introduction focusing on describing object-choice tasks and contrasting it with match-to-sample paradigms. References to children's production of drawing have been excluded for the most part.

R.3.4
I appreciate that the studies were pre-registered. Were the sample sizes and, in particular, the goal of testing 2 children per month between 3 and 7 years of age, based on power analyses? If yes, it would be helpful to note this in the manuscript.

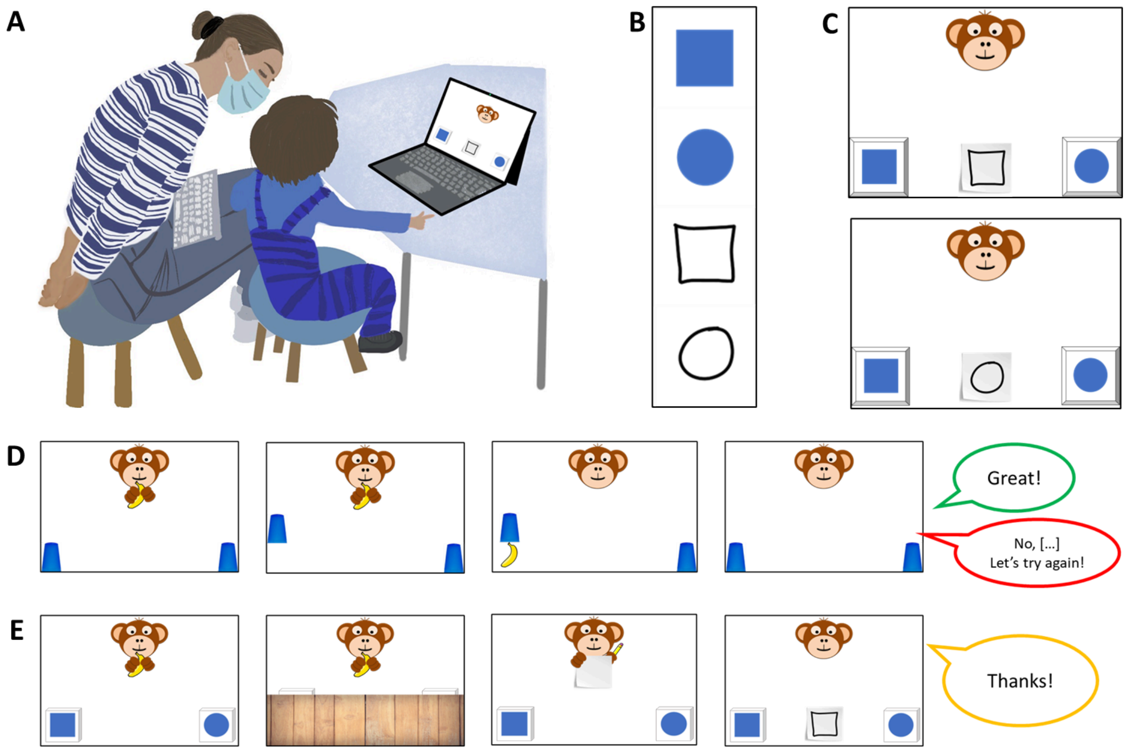
Response:
In the preregistration we stated:

> *Starting at 36 months, we will test two children per month over a range of four years (48 months) for a sample of 96 children. While using age as a continuous predictor is less common in developmental studies, this design yields a sample of at least 24 children per year of age which is in line with conventions in the field.*

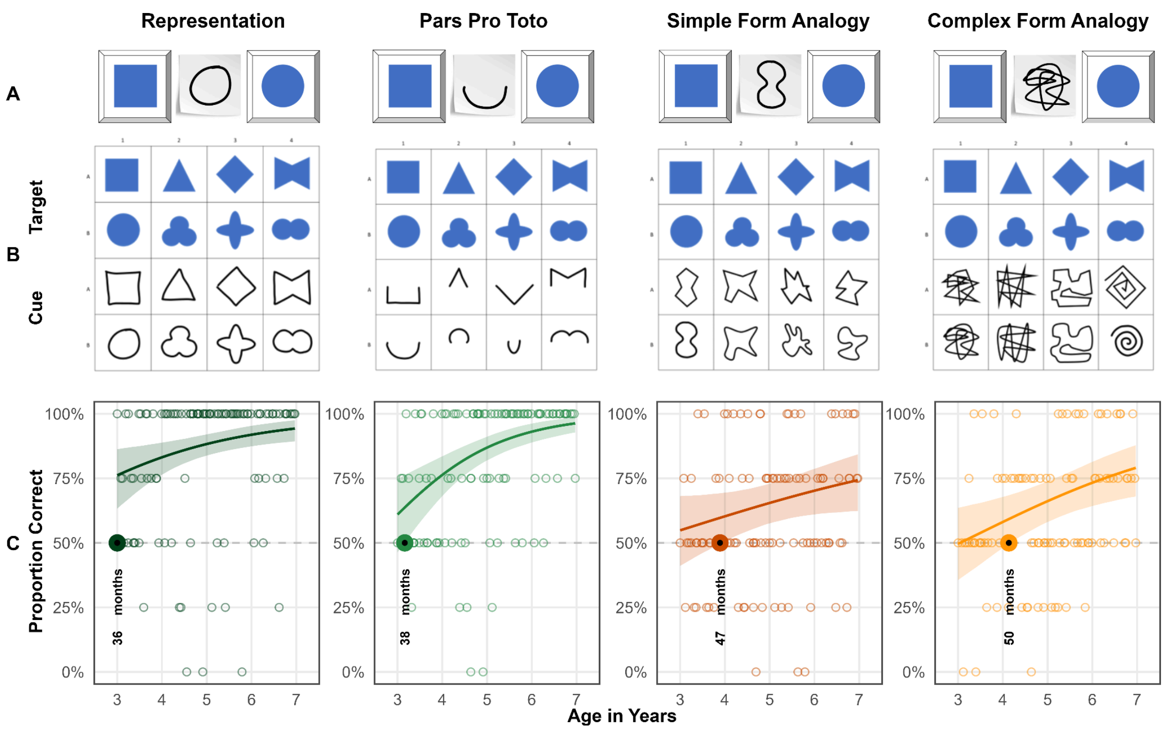
Our main aim was to collect a balanced continuous sample to determine the ages at which children succeed with the respective symbol-referent conditions. At the time of preregistration we did not have estimates for effect sizes in this paradigm and analytic approach, such that we opted for default priors and a sample aligning with conventions in the field.

R.3.5
I appreciate the figures depicting the results, however, it would greatly help the reader if there was a figure earlier in the methods depicting the different types of representations presented. It really helps understand what the child is seeing and what the researchers are testing.

Response:
The general methods section contains a figure (see below) that illustrates (A) the setup with which children were tested, (B) an example of cue and target stimuli of a specific trial and (C) how they were arranged to create the test slides. Finally, we show (D) how these would be inserted into a test trial.



We have opted to present all of the stimuli together with the results as this is particularly helpful when evaluating them. To guide the reader, we have added a reference to figures 2, 3, and 4 respectively when discussing stimulus design in the general methods section.



R.3.6

On p. 20, the authors indicate that "children perform above chance… as early as 36 months" . Later they state "children succeed only at 38 months in Pars Pro Toto". Given that the chance-level analyses are based on ages binned by years and only 2 children are tested exactly at each months, these statements are not quite accurate and should be revised. This comment applies throughout.

Related to the comment above, it is not quite clear which analyses the authors are using to determine the age at which children succeed—can this be made clearer? I am assuming it is the chance-level t-tests, as age does not interact in the models in Expt 1, except for the Pars pro toto condition. It is possible that I am not clear on how the Bayes models are being conducted.

Response:

The children are only binned according to their age in years for the additional conventional analyses in the supplements. In the supplements we provide conventional t-tests for the convenience of readers that are less familiar with the main analytic approach and may want to compare results. This is mentioned in the last paragraph of the methods section.

Our main analyses are Bayesian GLMMs that allow us to continuously model the likelihood of children making correct choices across the entire age-range from three to seven years of age. These analyses are not restricted by bins according to years of age. In the methods section we then also describe how we determine the age of success:

> *To answer the main research question of when children as a group systematically make correct choices in any of the conditions outlined below, we followed the approach of House and colleagues (House et al., 2013) and plotted the model estimates graphically as a smoothed curve surrounded by the respective confidence intervals. We fitted models to predict the developmental trajectory (with 95% CrI) of group level performance drawn from values of the posterior predicted distribution via the function fitted. These trajectories and CrIs were plotted by age. The criterion for settling when children performed above chance was the point at which the lower bound of the 95% CrI for a particular trajectory did no longer overlap with a mid-line demarcating the 50% chance level.*

Hence, we are able to provide very robust estimates of the age in months when group level performance exceeds chance level and this is central to the three studies presented. We made amendments to the manuscript to better guide the reader:

- We added a paragraph break when introducing how we determined the age of success. The paragraph opens with "To answer our main research question of when children perform above chance [...]"
- When referring to the additional exploratory t-tests in the supplements, we specifically highlight that these are only used to determine whether an age-bin is

above chance and are only added in the supplements for readers that might want to compare our main analyses with conventional t-tests.
- We now directly highlight the "age of success" when referring to the result figures of study 1, 2, and 3.

R.3.7

The Discussion section provides a detailed review of the results. I think this could be condensed, as it repeats what is noted earlier, and more attention could be devoted to the mechanisms that support children's reasoning in these studies (as noted earlier, I would suggest comparison and analogical reasoning be discussed in more detail).

Response:

We followed the suggestions of reviewer 3 to reduce redundancy by shortening the interim summaries following each study, as well as the final summary. We have introduced several new references and work by Gentner and others. We found a paper by Cooperrider and Goldin‐Meadow (2017) particularly fitting that discusses relations between gesture and analogical reasoning as it also connects the literature on analogical reasoning with a communicative phenomenon. The authors distinguish gestures capturing attributes with others capturing relations and refer to the latter as analogical gestures. This ties in with the object/feature distinction we use (i.e. size of object / size of feature) as well as the difference between. Our work generally ties in with basic tenets of Gentner and Hoyos (2017) such as "[...] the more different the two comparands, [...] the less likely children are to spontaneously compare [...] and align them."

Cooperrider, K., & Goldin‐Meadow, S. (2017). When gesture becomes analogy. *Topics in Cognitive Science*, *9*(3), 719-737.

Gentner, D., & Hoyos, C. (2017). Analogy and abstraction. *Topics in cognitive science*, *9*(3), 672-693.

Winter, B., Woodin, G., & Perlman, M. (2026). Defining iconicity for the cognitive sciences. In O. Fischer, K. Akita, & P. Perniss (Eds.), *The Oxford handbook of iconicity in language* (pp. 138–150). Oxford University Press.

R.3.8

The authors propose that the results signal a "qualitative shift" in reasoning. I would suggest that this conclusion needs to be better supported than it currently is. It is very difficult, based on the data provided, to distinguish a qualitative shift from quantitative growth. The figures depicting the developmental growth do not suggest abrupt or discontinuous changes.

Response:

We agree that our visualisations and results are strictly speaking not sufficient for separating a quantitative from a deep qualitative shift in the underlying cognitive processes. However, it is remarkable that across three studies with different samples and across a versatile set of conditions, we find evidence for success around the fourth birthday. We have replaced all instances of the phrase “qualitative shift” with “major shift” in order to draw the reader's attention to these converging results without specific implications as to the underlying cognitive developments.

Young Children's Spontaneous Comprehension of Symbol-Referent Relationships in the Graphic Domain

For Review Only

Abstract

Young children's communicative skills in language and gesture are well documented, yet much less is known about their ability to interpret graphical symbols. Across three coordinated studies using identical procedures, we examined children's spontaneous comprehension of unfamiliar graphic symbols in a simple object-choice task spanning a broad range of symbol–referent relationships. Each study included four conditions in which reference was established through direct representation, part–whole relations, analogies in size, number, and form, or gestalt principles such as orientation, alignment, and figure–ground organization. Participants were 304 German-speaking children continuously sampled from age 3 to 7 (Study 1, N = 106, 51 female; Study 2, N = 99, 49 female; Study 3, N = 99, 55 female). Children reliably used direct representations to identify target objects before age 3, whereas group-level success in the more abstract conditions emerged later but consistently before school entry. Converging evidence across conditions involving abstract or conceptual correspondences indicates a major shift in symbolic competence around age 4. Together, these studies trace children's developing ability to understand graphical reference across a continuum from iconic to abstract and offer one of the most systematic investigations to date of how young children derive meaning from unfamiliar graphic symbols.

Keywords: Graphic Communication, Pragmatics, Symbol-Referent-Relationships, Iconicity, Representation, Analogical Reasoning

Word count: 12.420

For Review Only

Young Children's Spontaneous Comprehension of Symbol-Referent Relationships in the Graphic Domain

Lay Summary (max. 120 words)

Young children are skilled at understanding spoken language and gestures, but much less is known about how they make sense of graphic communication. We explored how children aged 3 to 7 interpret unfamiliar graphic displays in a picture-book-style game of hide and seek where they had to choose the correct hiding place based on a drawn cue. Across three studies, we tested many kinds of symbol–object relationships, ranging from clear representations to abstract cues based on shape, size, number, orientation or spatial arrangement. Children followed representational drawings at three, but the understanding of more abstract symbols emerged around the fourth birthday. These findings show how children gradually learn to find meaning in increasingly complex visual symbols even prior to literacy.

Introduction

The development of symbolic understanding - encompassing language, gesture, and symbolic artifacts - is a cornerstone of human enculturation and a hallmark of the cognitive capacities defining the human mind (Tomasello, 2014; Vygotsky, 1980). Within this developmental trajectory, the acquisition of oral language and the subsequent mastery of literacy (Bruner, 1996; Dehaene, 2009; Olson, 1994) represent two of the most transformative milestones in cultural learning, each yielding profound consequences for children's cognitive architecture (Karmiloff-Smith, 1994). To elucidate the mechanisms underlying these shifts, researchers have examined how children from infancy through the preschool years negotiate various symbolic

For Review Only

media, including gestures, words, photographs, representative drawings, maps, models, and sound symbolism (Callaghan & Corbit, 2015).

The symbolic object-choice task serves as a primary paradigm for investigating these capacities. In a typical procedure, children engage in a "hide-and-seek" game where, after familiarization with the goal of retrieving a target from one of two containers, the hiding process is obscured. At test, the experimenter provides a communicative cue to guide the search. Researchers have manipulated these cues to probe the ontogeny of symbolic appreciation, ranging from social signals like gaze, reaching, pointing or iconic gestures (Behne, Liszkowski, Carpenter, & Tomasello, 2012; Bohn, Call, & Tomasello, 2019; Sodian & Thoermer, 2004; Tomasello, Carpenter, & Liszkowski, 2007; Tomasello, Striano, & Rochat, 1999), to more abstract or iconic referents such as arrows, markers, replicas (Callaghan, 2000; Leekam, Solomon, & Teoh, 2010; Tomasello, Call, & Gluckman, 1997), graphic symbols such as drawings or photographs (Callaghan, 2000; DeLoache, 1991), and even arbitrary signals like turning on a light (Moore, Liebal, & Tomasello, 2013). Standard analyses of this paradigm (Tomasello, 2010) suggest that success requires navigating several inferential layers. Within a cooperative framework, the child must first construe the interaction as a joint endeavor with a competent partner, allowing them to treat the cue as a deliberate communicative act rather than a random event. Retrieval then necessitates a dual inference: mapping the symbol-referent relationship - whether via deixis, iconicity, or spatio-temporal contiguity - while simultaneously recognizing the cue's pragmatic relevance to the immediate goal. Ultimately, the task demands both a structural match between cue and target, and a pragmatic inference regarding the cue's symbolic intent.

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Although infants grasp naturally meaningful behaviors like pointing and gaze by the end of their first year, their ability to exploit less transparent symbolic artifacts emerges significantly later. The seminal work of DeLoache and colleagues has been instrumental in charting this progression, demonstrating that children's capacity to use scale models, pictures, videos, and maps as informative tools undergoes a rapid transformation during the third year of life (DeLoache, 1991, 2004, 2011). A robust finding across this literature is the 30-to-36-month transition: while 30-month-olds typically fail to use a scale model to locate a hidden item, 36-month-olds excel, mastering the concept of "dual representation" enabling them to see an artifact as both an object and a symbol (Callaghan & Corbit, 2015; DeLoache, 2004). A parallel developmental trajectory is evident in the domain of graphic representation: functional use of graphic representations such as line or pencil drawings in choice tasks remains fragile until approximately age three. For instance, while children as young as 30 months may use shape to interpret drawings when intentionality is clear (Gelman & Ebeling, 1998), Callaghan (1999) found that at 24 months children still struggle to select items based on representative drawings. This fragility is underscored by findings that 30-month-olds often fail to benefit from iconicity in either drawings or replica objects, whereas 36-month-olds reliably succeed in object-choice-tasks across varying levels of stylization (Callaghan, 2000; DeLoache, 1995).

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Hence, from 36 months onward, it may be productive to investigate whether children can comprehend symbol-referent-relationships that are less dependent on iconicity. Any symbol-referent-relationship that children can spontaneously use early may be of particular relevance for our understanding of emerging and graphical literacy and the development of graphic systems of communication. While iconicity provides an entry point for symbol-grounding in graphic communication (Callaghan, 2000) much like in the gestural domain (Bohn, Call, et al., 2019; Callaghan & Corbit, 2015; Fay, Ellison, & Garrod, 2014), it is generally also possible to establish reference via more abstract or conceptual symbol-referent relationships such as size, number, orientation, position or gestalt principles serving as fundamental avenues for encoding information in the graphic domain (Cleveland & McGill, 1984; Donnelly et al., 2007; Stevenson, Alberto, Boom, & Boeck, 2014). Defining the precise ages and relative developmental trajectories for children's sensitivity to using these graphic primitives as conveying meaning and for disambiguating reference in object-choice-tasks will provide crucial insights into children's developing symbolic competence and the interplay of communicative and analogical capacities.

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To explore the continuum from iconic to abstract symbolic relationships, researchers can systematically reduce the resemblance between cue and target. Rather than varying levels of pictorial realism (Callaghan, 2000), one can minimize similarity to isolate the conceptual foundations of the mapping. While using such "reduced" cues may appear to violate the Gricean maxims of quantity (providing sufficient information) and manner (avoiding obscurity) (Grice, 1991), children are well-accustomed to suboptimal communication in daily life. Through processes like streamlining and conventionalization, they frequently encounter and successfully navigate ambiguous utterances; indeed, they often master and enjoy the interpretive decoding required in communicative games like charades (Bohn, Call, et al., 2019). An interesting case in question may be Pars Pro Toto in the graphic domain: part-whole relationships on the level of geometric figures could be an ideal target for an investigation into slightly abstract symbol-reference relationships as shape recognition and naming are robust in the third and fourth year (Verdine, Bunger, Athanasopoulou, Golinkoff, & Hirsh-Pasek, 2017; Verdine, Lucca, Golinkoff, Hirsh-Pasek, & Newcombe, 2016; Zambrzycka, Kotsopoulos, Lee, & Makosz, 2017) and the ability to mentally complete canonical geometric shapes is already nascent in early infancy (Kellman & Spelke, 1983; Valenza, Leo, Gava, & Simion, 2006), yet there is currently no data on when children may be able to use pars pro toto cues in an object-choice context.

Another way of reducing iconicity may be the metaphorical use of form, such as using sharp or spiky drawings to refer to more square or round shapes (Knoeferle, Li, Maggioni, & Spence, 2017). This phenomenon is central to research on sound symbolism within the lexicon (Bermúdez, 2020; Sidhu, 2025), as well as in studies of action and gesture (Bohn, Call, et al., 2019; Margiotoudi & Pulvermüller, 2020). Robust across cultures (Ćwiek et al., 2022), this sensitivity emerges around 30 months for word-shape matching (Fort et al., 2018; Imai, Kita, Nagumo, & Okada, 2008; Maurer, Pathman, & Mondloch, 2006) and by 36 months for matching actions with gestures (Bohn, Call, et al., 2019). Despite these findings, it remains unknown whether children can utilize the global properties or “impressions” of a shape’s form to ground reference in the graphic domain alone.

For Review Only

To minimize iconicity in symbol-referent relationships, cues and targets may share global properties like orientation and alignment rather than specific shapes. Children demonstrate an early sensitivity to these compositional principles in their drawings, maintaining realistic spatial configurations (e.g. placing wheels beneath a car when drawing) and avoiding figure overlap (Cox, 2005; Luquet & Costall, 2001; Piaget, Inhelder, & Bovet, 1997). However, there is virtually no data on how likely children are to use Gestalt principles as a basis for forming symbol-referent relationships, except for the use of relative position in the developing ability to read maps. While four-year-olds possess an intuitive grasp of relational directions (e.g., “behind” or “in front”) in basic maps (Landau, 1986), iconicity remains an important scaffold for their conceptual understanding (Liben & Downs, 2013). Furthermore, although preschoolers can use geometric map information for local navigation (Shusterman, Ah Lee, & Spelke, 2008), the ability to transfer these relational cues to large-scale spatial arrays does not stabilize until age six or seven (Blades & Spencer, 1994; Uttal & Wellman, 1989).

Finally, alignment between cue and target may also be based on analogies in size and number disregarding shape. Size is a relative property of graphical content and requires a comparison, for example when drawing a big and small object or conveying size by making an object occupy a large part of the frame of reference (Cox, 2005). Children are able to make spontaneous or productive size comparisons at three years of age and begin to use size words (*big*), size superlatives (*biggest*), or quantifiers (*more*, *most*) (Ferry et al., 2025), yet young children are most accurate with *bigger* and only gradually gain competence with words like *smaller*, *shorter* or *longer* (Frausel et al., 2020). When presented with an array of graphic stimuli, preschoolers can reliably point at, for example, the smallest or largest item in a set (Haun & Tomasello, 2011). One of the few studies using size as a way for grounding reference, investigated the role of the producers intentions in identifying concepts in ambiguous scribble drawings that were presented to participants as aiming to depict, for example, a small spider next to a big house. Children here made use of spatial analogies beginning at three years of age but appeared to benefit more as they turned four (Bloom & Markson, 1998). Similarly, number is a relational primitive and a foundational element of graphic communication (Cooperrider & Gentner, 2019; Ifrah, 2000). Sensitivity to number per se develops from an infant's implicit grasp of proximate numerical differences (Carey, 2000; Starr, Libertus, & Brannon, 2013; Xu & Arriaga, 2007) to an explicit understanding of numerical systems (Carey & Barner, 2019). Supported by early home environments (Daucourt, Napoli, Quinn, Wood, & Hart, 2021) and preschool curricula (Kucharz, 2012; Schmidt, Sauerbrey, & Smidt, 2021), children typically master the cardinality principle by age four (Carey & Barner, 2019; Silver & Libertus, 2022). Preschoolers are well equipped to successfully apply their implicit and explicit numerical knowledge in practical contexts, such as comparing magnitudes smaller than five (Wynn, 1990).

For Review Only

While children appear generally sensitive to and able to make comparisons along the conceptual dimensions of shape, orientation, size and number, studies directly testing them as grounds for establishing reference are very rare. A highly valuable point of reference can be found in the literature on relational and non-relational match-to-sample tasks where studies often comprise various stimulus sets or conditions probing these matching dimensions. While matching shapes in match-to-sample (MTS) tasks poses little challenge for children beyond age three (Gerhardstein & Rovee-Collier, 2002; Kroupin & Carey, 2022b), utilizing relational dimensions like size or number remains difficult. Even by age four, children rarely spontaneously infer size as a matching criterion in three-item MTS arrays, and success in number-based matching typically emerges between ages four and five (Kroupin & Carey, 2022b). These difficulties intensify in relational match-to-sample (RMTS) tasks, which require coordinating at least six items to identify higher-order patterns of identity (AA, BB, AB) or difference (AB, BA, BB). Mastery across dimensions is generally absent prior to age five, as the task requires navigating abstract generalizations and suppressing prepotent responses to superficial features - taxing both executive function and working memory (Gentner & Hoyos, 2017; Kroupin & Carey, 2022a, 2022b). However, performance can be scaffolded. Four-year-olds succeed when relations are embedded in causal narratives (Goddu, Lombrozo, & Gopnik, 2020), and even three-year-olds improve following explicit training with the terms *same* and *different* (Christie & Gentner, 2014). A critical distinction exists between these paradigms and symbolic object-choice tasks. In MTS/RMTS, children are explicitly instructed to find matches as an end in itself. Conversely, the symbolic object-choice context requires an additional communicative inference: children must recognize that a cue was produced intentionally to guide their behavior and that its meaning derives from a specific relation to a target. Consequently, symbolic object-choice tasks represent

For Review Only

a more demanding and ecologically valid testing ground, revealing whether young children can deploy their nascent analogical reasoning skills within a practical, goal-directed interaction.

From this overview, several research desiderata emerge. The prevailing literature's heavy reliance on similarity and iconicity in the context of graphic communication (Callaghan & Corbit, 2015) leaves the broader scope of children's symbolic flexibility largely unexplored. Because most studies employ depictions of real-world objects, results may be influenced by children's familiarity with canonical features (Long, Moher, Carey, & Konkle, 2019) or cultural graphic conventions. The present work provides a systematic investigation into the diverse conceptual dimensions along which graphic symbols can be imbued with meaning and traces development continuously through the pre-school years employing the symbolic object-choice task as one coherent and simple paradigm. By utilizing novel abstract graphic displays as cues and geometric shapes as targets, this study isolates children's ability to identify conceptual features and apply pragmatic inferences in context, largely independent of prior knowledge of script, numerals, or conventional icons. Furthermore, by omitting training, instruction, or feedback, our approach ensures that responses reflect what children find naturally meaningful. Consequently, the studies below showcase young children's capacity to spontaneously grasp novel symbol-referent pairings, providing critical insights into the learnability and potentially the emergence of communicative systems during a developmental period of increasing conventional symbolic mastery.

General Methods

All studies presented below share methodological and analytic approaches. For the convenience of the reader, common features of the procedure, participant recruiting and stimulus

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design are reported first before discussing the three studies respectively. All studies were preregistered online prior to data collection (cf. Study 1: https://aspredicted.org/SJT_H7F, Study 2: https://aspredicted.org/L2H_XC7, and Study 3: https://aspredicted.org/DR4_B4B). All stimuli, experiments, data and analyses are available at: [MASKED FOR REVIEW]

Data Collection and Setup

In order to continuously trace the development of symbolic competences across the preschool years, for each study we aimed to test a minimum of two children per month of age between the third and the seventh birthday for a total of 96 participants. The final sample approximates an equal distribution of male and female participants and aligns with conventions in the field in providing a minimum of 24 participants per study and year of age. As children participated on the basis of availability and data were collected by several experimenter teams in parallel, the number of participants slightly exceed the preregistered minimum sample size. For an overview of the distribution of all children included in the analyses across the age-range see supplement figure 1. Figure 2 from the supplements shows the age-distribution of all children that participated but were not submitted to analyses due to the exclusion criteria outlined below. All participants were recruited in Leipzig, a medium-sized European city, and came from a predominantly white population of middle to high income families. Children were largely tested in day- and after school-care, and occasionally in the lab or at home. Data collection took place from June 2022 to February 2023.

For Review Only

Setup and Procedure

During test sessions, a child and an experimenter sat down together to play a picture-book-style hiding game presented on a touch-screen laptop. Verbal instructions were played back by the experimental script. Experimenters supervised children and occasionally reacted with a fixed set of verbal prompts when children were not following the presentation ("Oh look, the game continues!"). See figure 1 A for an illustration of the setup.

Familiarization. First, the experimental script introduced a cartoon monkey. This character placed two cups on the bottom left and right side of the screen. After holding up a banana, one of the barriers was lifted, the banana was placed underneath one of the cups and the barrier was lowered. Children were now prompted to touch the hiding place and in doing so the barrier of their choice was lifted to reveal the banana if they chose correctly. The experimental script provided feedback upon children's choice ("yes, great job!"; "No, that's not it. Let's try again!") during the familiarization (cf. figure 1 D). To ensure that children were familiar with the goal of the game and the touch interface, they first had to complete a set of four to eight familiarization trials with a success rate of 75%. In case a child did not reply correctly in three out of four trials, another four familiarization trials were provided. If the child was correct in six out of eight trials, she was included in the main sample. Children that did not succeed were allowed to participate but their data was not submitted to analysis.

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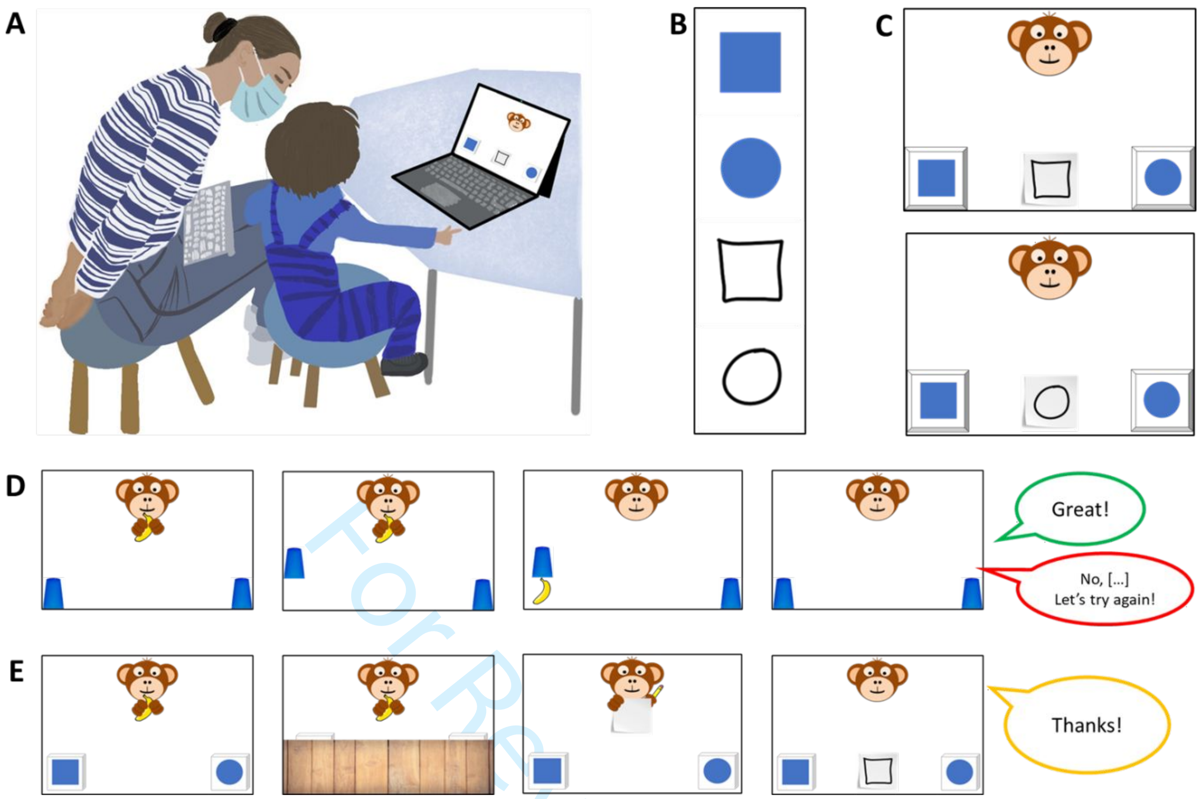

Figure 1. *Setup, Stimuli and Trial Structure.* Panel A shows the setup during testing. Experimenters were sitting behind the children in order to not distract them while supervising data collection. Panel B shows all elements contributing to an item of a possible test trial consisting of two choice options and one of two possible cues. Panel C exemplifies how two choice options and one of the cues were arranged for a specific test trial. Panel D shows a familiarization trial and Panel E shows a test trial.

For Review Only

Test. The main phase of the study commenced with announcing that the cartoon character had an idea for a new game. Now, children were not allowed to see where the banana would be hidden, but that the monkey would help them find it. The cartoon character was established as a knowledgeable and benevolent partner in a cooperative coordination game ("Coco is going to help you find the banana. Coco is drawing something for you."). After the hiding phase, the monkey held up a piece of paper and a pencil. Pencil movement and a short scribble sound indicated that the monkey was drawing. Finally, participants were prompted with the phrase "Where is the banana?" and the monkey's drawing was placed in the center between the two hiding places. The drawing served as a cue to guide children's choice. Children's choice was acknowledged with neutral feedback ("Ah, thank you") leading to the next trial (cf. figure 1 E).

A single trial lasted roughly 20 to 60 seconds, depending on how swiftly children chose. Each study presented four different experimental conditions with four trials each in a blocked order for a maximum of 16 test trials. Unique combinations of graphic displays were used on every trial. Test sessions lasted about 12 minutes in total. The order of conditions was counterbalanced across participants. Children occasionally wished to stop before completing all trials, resulting in minor deviations of the total number of trials per condition that were submitted to analysis. For an overview of the average number of trials participants received in each condition, see supplementary materials. In line with the preregistration, children had to complete a minimum of eight test trials to be included in analysis.

Stimuli and Counterbalancing

The context in the studies presented below is provided by a game of hide and seek and the options are two hiding places that are distinct by means of the graphic displays they are marked with. The aim was to test at what age children become able to spontaneously match cue and target along various dimensions of symbol-referent relationships. For this, the graphic displays presented as choice options were designed to saliently differ in one relevant dimension and to be as similar as possible with regard to other features. The cue, on the other hand, was a reduced and less straight-lined graphic display akin to a hand drawing that had something in common with one of the choice options and was thereby referring to it. In test trials, one of the choice options serves as a target and the other as a distractor. For counterbalancing, a second cue was designed

For Review Only

to refer to the other choice option, such that across participants the same choice options serve equally often as targets and distractors. For an illustration of trial composition, see figure 1 B and C. For each of the conditions in the three studies below, four sets of stimuli were designed, consisting of two blue shapes serving as target or distractor, and two drawings that could serve as cues. Each condition covers a particular type of symbol-referent relationship via four stimulus versions with two variations each. For an example, consider figure 2. The first box in Panel B shows all stimuli for the *Representation* condition. The first column exemplifies a set of targets (a blue square and circle) and cues (outline drawings of a square and circle). During test, participants are presented with four test trials per condition, each composed of the shapes of a single column. A child sees each trial combination only once and with only one of the two possible cues. Across children, the position (left/right) of the target, and the identity of the cue are counterbalanced. For the convenience of the reader, all stimuli employed across studies are presented with the results for each condition in figures 2, 3, and 4 respectively.

For Review Only

Data Handling and Analyses

Upon touching the screen, children's choices were logged and immediately coded as correct or incorrect by the code running the stimulus presentation. To evaluate how children's ability to interpret cues varies by age and condition, we employed Bayesian Logistic Generalized Linear Mixed Models (GLMMs). This mode of analysis is appropriate as it allows to estimate the probability of success while accounting for uncertainty and individual differences, and because it allows mapping linear predictors to a probability between 0 and 1 via a link function. Our analyses modeled children's binary choices (correct vs. incorrect) using fixed effects for condition and age, as well as their interaction as primary predictors. In addition, we included a fixed effect of trial number, to control for learning or fatigue, and a fixed effect of sex. To account for the possibility that some children may be better at the task than others, we included random intercepts for each participant. We also included random slopes for trial number, allowing potential effects of learning or fatigue to vary from child to child. To facilitate processing and interpretability, age and trial number were standardized to a natural mean of 0 prior to modeling. The full model notation was `correct ~ condition*z.age + z.trial + sex + (z.trial | subid)`. In line with best-practice conventions, we evaluated the relevance of age and condition for children's performance prior to interpreting coefficients. For this, the full model outlined above was compared with a reduced model lacking the interaction of age and condition. To assess the fit and relative explanatory power of either model, we used Widely Applicable Information Criterion (WAIC) scores and weights (McElreath, 2018) as well as the difference in Expected Log Predictive Density (ELPD) via the function *loo_compare* (Sivula, Magnusson, Matamoros, & Vehtari, 2020). Across all studies, the preregistered full model proved superior or equally predictive. Model estimates were inspected for the different predictors including their 95% Credible Interval (CrI). In each study, the condition hypothesized as the most simple was set as the reference level within conditions. All models were implemented in Stan (Stan Development Team, n.d.) via the function *brm* of the package *brms* (Bürkner, 2017).

In order to answer our main research question of when children systematically perform above chance in each condition, we followed the approach of House and colleagues (House et al., 2013) and plotted the model estimates as a smoothed curve surrounded by Credible Intervals (CrI). We generated developmental trajectories with 95% CrIs from the posterior predicted distribution via the function *fitted* and plotted them with children's age on the horizontal axis and the probability of correct choices on the vertical axis. Chance level was marked via a 50% chance line for orientation. Children as a group were considered to perform above chance only when the

For Review Only

lower bound of the 95% CrI no longer overlapped with the 50% chance line (e.g. figures 2, 3, and 4).

All analyses and our hypotheses about the relative difficulty of conditions were preregistered prior to data collection. However, the specific age-point predictions are exploratory, and our use of ELPD for model comparison (Sivula et al., 2020) represents a modern refinement over the methods available at the time of preregistration. To additionally evaluate variability at the level of the items tested, additional exploratory analyses included item level random effects (Model notation `correct ~ condition*z.age +z.trial +z.sex +(z.trial|id) +(z.age|item)`). For brevity, tables outlining the coefficients for the final models were moved to the supplementary materials along with the exploratory item-level analyses outlined above. The supplements also provide conventional analyses binning participants according to their age in years for readers that are less familiar with the analyses described above. In the supplements, two-tailed one-sample t-tests with a chance level set to .5 were computed for participants binned according to year of age. Here, Cohen's *d* is provided as a standardized effect size for significance testing.

For Review Only

Study 1

Study 1 aimed to establish a baseline for children's performance and for evaluating task demands by providing the simplest symbol-referent relationship possible, where the cue is a direct representation of the target (*Representation*), which children should master at the end of the third year (Callaghan, 1999, 2000). From this, three further conditions were derived that were also employing form or shape as a means for establishing reference but that were less iconic by either reducing the amount of information provided (*Pars Pro Toto*), or the amount of similarity between cue and target (*Simple Form Analogy, Complex Form Analogy*). With *Pars Pro Toto* likely being more challenging than *Representation* due to the completion required, and form analogies being relatively abstract, our preregistered hypothesis was that as a group, children will first succeed with *Representation*, then *Pars Pro Toto*, *Simple Form Analogy* and finally *Complex Form Analogy*.

Stimuli

To make the four conditions in study 1 as comparable as possible, they are all employing the same target stimuli (cf. figure 2, panels A and B). In addition, the two cues within a trial can be seen as the round and square equivalents of each other. In *Representation,* the graphical cue is an outline drawing of the target. The second condition, *Pars Pro Toto,* refers to the targets by means of a part-whole relationship. This requires children to complete the shapes according to gestalt principles. While this is easiest in the first two items of *Pars pro Toto* due to the canonical shapes (square, circle) it is less obvious with the less canonical shapes albeit they are vertically and horizontally symmetrical on purpose (cf. 2 B, box 2). For comparability, *Pars Pro Toto* uses the same graphical cues as *Representation* but cut in half at a horizontal mid-line. A Stimulus set of an individual trial either uses the top or bottom half, but both variations are counterbalanced across trials. Two further conditions aimed to abstract from the original representational symbol-referent relationship by providing graphical analogies in form. In both *Simple Form Analogy* and *Complex Form Analogy* the cue was an abstract line drawing being more round or rectangular, thereby referring to either the round or rectangular equivalent of the target shapes. As this has not been done before in developmental research, our aim was to provide two versions of form analogies both supporting children's comprehension in distinct ways. In *Simple Form Analogy*, the cues are less dense and therefore simpler to grasp, whereas the more complex versions in

For Review Only

Complex Form Analogy provide more information. For an overview of all stimuli employed in Study 1, see figure 2 B.

Participants and Sample

A sample of 106 children (M = 59.18 months, SD = 13.58 months, range 36 - 83 months; 51 female) participated in Study 1. In addition, 22 children (11 female) were tested but excluded from analysis for not succeeding during familiarization (N = 13), for not completing at least eight out of 16 test trials (N = 1), or due to being fussy (N = 2). For 4 children, experimenters only learned during testing that they were not fluent enough in German to participate as their families had only recently migrated. Finally, 2 children had to be excluded due to technical issues. A total of 1688 trials (mean per condition = 422, range: 420 - 424) from 106 participants were submitted for analysis.

Results

Posterior predictive checks (PPC) for both full and null model indicated excellent fit of observed data and model predictions. Rhat values in both models were equal to one, indicating convergence across all chains. Effective sample sizes for all fixed effects in the full model (Bulk ESS, mean = 6222, range 4323 - 10098) and the null model (Bulk ESS, mean = 5777, range 4351 - 8307) were > 1000, indicating reliable posterior estimations.

Comparing the models using weights based on the Widely Applicable Information Criterion (WAIC) yielded 74.52% of the model weight for the full model, and 25.48% for the null model. Hence, the full model generally had a higher probability of making accurate predictions. Directly comparing the models' WAIC via expected log predictive density (ELPD) corroborates this (ELPD WAIC; full model = -901.11; null model = -903.53). The standard error

For Review Only

of the difference in predictive accuracy (SE = 3.14) is not much larger than the difference itself (ELPD diff = -2.42). Hence, evidence in favor of this model is not decisive. A similar comparison via Leave-One-Out Cross-Validation (LOO) provided essentially the same results. In absence of conclusive evidence for either model, we report the results for the full model in line with the preregistration.

Relative to the *Representation* condition, the *Simple Form Analogy* (β = -1.39, 95% CrI [-1.75, -1.03]) and *Complex Form Analogy* (β = -1.36, 95% CrI [-1.73, -1.00]) have a considerably lower probability of correct responses. The *Pars Pro Toto* condition has no clear difference from the *Representation* condition (β = -0.14, 95% CrI [-0.54, 0.28]). Interaction effects with age were not relevant with the exception of *Pars Pro Toto*. Here, the interaction with age was positive and just above zero (β = 0.30, 95% CrI [-0.05, 0.67]), suggesting that performance increased more steeply across the age range than in the *Representation* condition. Generally, participants' performance improved with age in all conditions (β = 0.42, 95% CrI [0.13, 0.72]). In contrast, trial number has no clear effect on performance (β = -0.01, 95% CrI [-0.15, 0.12]), suggesting no evidence for learning or fatigue throughout the test session. For a side-by-side comparison of the developmental trajectories in the four conditions of study 1 and the respective ages of success, see figure 2.

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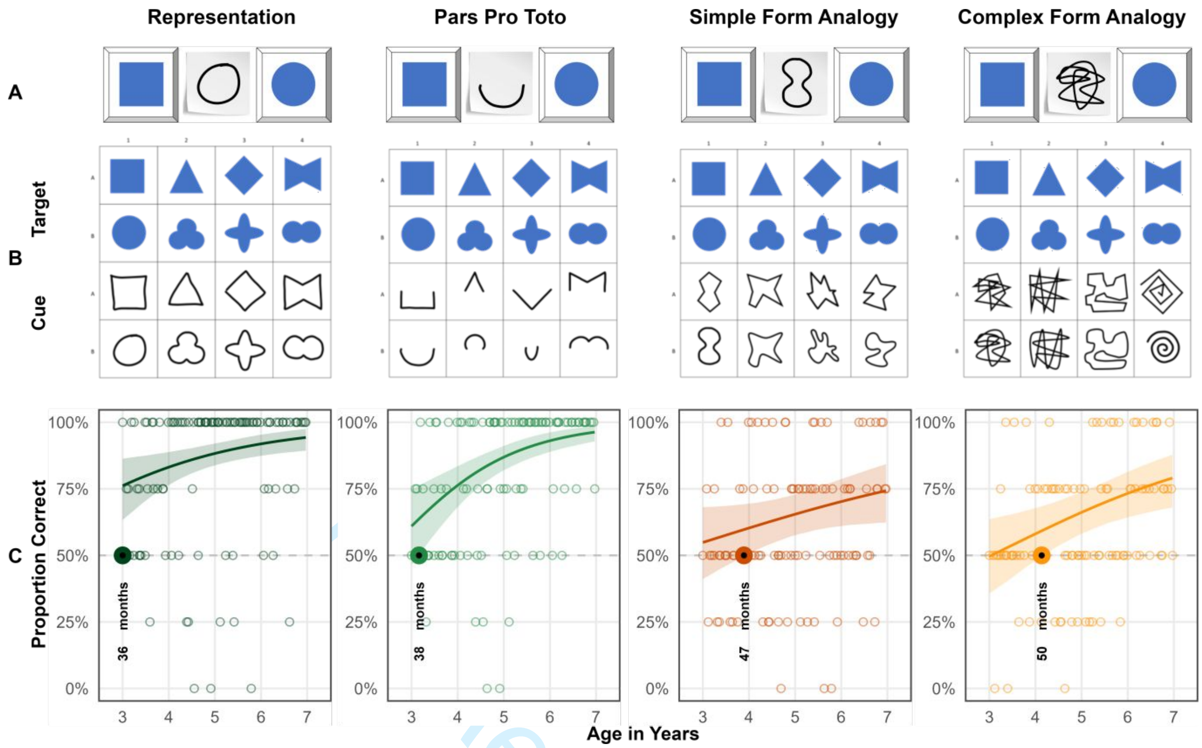

Figure 2. *Stimuli and Developmental Trajectories for Study 1.* Panels illustrate (A) an example stimulus combination (distractor, cue, target), (B) all items of the respective conditions and (C) results. Coloured lines indicate smoothed mean performance by age. Shaded areas represent 95% CIs. The dashed line demarcates chance level and the dots represent individual means. The colored dots and annotation indicate when children's performance exceeds chance level.

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Discussion

Children perform above chance in the *Representation* condition at least as early as 36 months, which is the lower limit of the age-range. Based on the literature, it is reasonable to assume that also in our setup, children would be able to solve the task at hand between the second and third birthday (Callaghan, 1999, 2000; DeLoache & Marzolf, 1992). This result establishes that the task design and setup are sufficiently clear even for the youngest children in the sample. Children succeed only at 38 months in *Pars Pro Toto*. The interaction with age shows that performance improves more quickly with age and that the area covered by the upper and lower CrI bounds shrinks considerably across the age range. In the more abstract conditions *Simple Form Analogy* and *Complex Form Analogy*, preschoolers meet criterion at 47 and 50 months respectively. While both are still based in iconicity, in so far as their interpretation involves a sense of resemblance with regard to their shape (Winter, Woodin, & Perlman, 2026), they are conceptually demanding at the same time. With regard to the influence of surface features, group level success occurs slightly earlier with the reduced cues employed in *Simple Form Analogy* (Richland, Morrison, & Holyoak, 2006). Finally, the results are perfectly in line with the hypothesized developmental succession of group level success coming to pass first with *Representation*, then *Pars Pro Toto*, *Simple Form Analogy* and *Complex Form Analogy*.

For Review Only

Study 2

The conditions in study 2 continue to abstract from a representational symbol-referent relationship by using different shapes for both cue and target stimuli. The conditions here, draw on gestalt-principles like figure-ground-relationship, continuity and parallelism. The symbol-referent-relationship in *Absolute Position* is established by the respective cue and target sharing the same position with regard to the reference frame they occur in. In *Relative Position* cue and target are each composed of two shapes. Reference is established by the respective shapes being closer or further apart or sharing a position on a horizontal or vertical axis. For *Orientation of Object*, symbol and cue are aligned along the same axis. The final condition, *Orientation of Feature* employs cue and target shapes with a salient feature that is either oriented up- or downward. Because processing whole objects is generally less demanding than isolating constituent features, and because the position conditions require coordinating relations between objects or reference frames (Gentner & Hoyos, 2017; Halford, Wilson, & Phillips, 1998;

Rattermann & Gentner, 1998; Richland et al., 2006), our preregistered hypothesis predicted a developmental hierarchy with success first in *Orientation of Object*, followed by *Orientation of Feature*, *Absolute Position*, and finally *Relative Position*.

Stimuli

For comparability, *Absolute Position* and *Relative Position* employed the same distinct shapes as cue and target respectively (circle vs. square, cross). In *Absolute Position* target positions were placed in top and bottom, or central and peripheral conditions in half of trails. For *Relative Position*, two trials present target stimuli that are closer together or further apart, as well as another two trials with target components being aligned on a horizontal or vertical axis. To ensure that cue and target are maximally distinct in all dimensions, cues and target items do not align on the same axis. To convey a sense of direction in *Orientation of Object,* cues and targets feature elongated shapes that are aligned either on a horizontal, vertical, or diagonal axis with rising or falling slope. The shapes are distinct oblong ellipses and rectangles or abstract shapes with more or less round or square features. For *Orientation of Feature*, targets and cues are either circles or squares with a salient feature like an opening or a dent. For an overview of all stimuli used in Study 2, see figure 3.

For Review Only

Participants and Sample

A total of 99 three- to seven-year-old children (M = 60.04 months, SD = 13.69 months, range 36 - 83 months; 49 female) participated. None of these children participated in study 1. In addition, a total of 13 children (6 female) were tested but excluded from analysis for failing familiarization (N = 9), being fussy (N = 1), not being fluent enough in German to follow the instructions (N = 1), or due to technical issues (N = 2). A total of 1561 trials (mean per condition = 390.25, range: 388 - 393) from 99 participants were submitted for analysis.

Results

For both full and null model, PPCs indicate excellent fit of observed data and model predictions. Rhat values in both models were equal to one, indicating convergence across all chains. Effective sample sizes for all fixed effects in the full model (Bulk ESS, mean = 6149, range 3485 - 7545) and the null model (Bulk ESS, mean = 6877, range 3839 - 10040) were > 1000, indicating reliable posterior estimations.

To compare model performance, we evaluated the WAIC estimates. The null model showed a slightly better predictive performance (ELPD = -888.15) compared to the full model (ELPD = -889.99). WAIC values also indicate better performance of the null model (WAIC = 1,776.30) over the full model (WAIC = 1,779.99). However, the difference between models (ELPD Diff = -1.84) falls within the bounds of uncertainty (SE = 1.76), suggesting no advantage in predictive accuracy. Hence, the preregistered analyses using the full model is reported below.

Across all conditions, performance improved with both age (β = 0.40, 95% CrI [0.14, 0.68]) and slightly with trial number (β = 0.26, 95% CrI [0.10, 0.43]). Relative to *Absolute Position*, children were generally less likely to correctly solve *Relative Position* (β = -0.28, 95% CrI [-0.64, 0.08]). Performance in *Orientation of Object* (β = -0.10, 95% CrI [-0.46, 0.27]) and *Orientation of Feature* (β = -0.12, 95% CrI [-0.48, 0.23]) was not substantially different from the reference category *Absolute Position* when considering the full age range. Interaction terms between age and condition were not credibly different from zero, suggesting similar developmental patterns for all conditions. For a side-by-side comparison of the developmental trajectories and the respective ages of group-level success, see figure 3.

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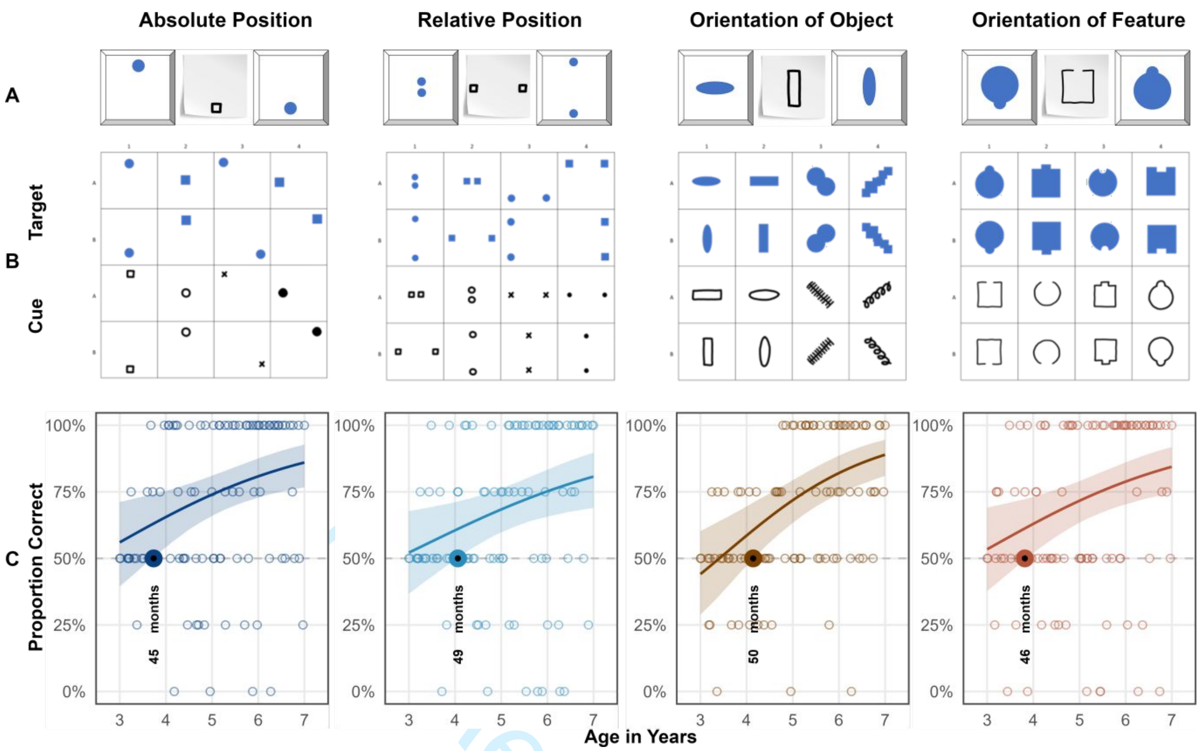

Figure 3. *Stimuli and Developmental Trajectories for Study 1.* Panels illustrate (A) an example stimulus combination (distractor, cue, target), (B) all items of the respective conditions and (C) results. Coloured lines indicate smoothed mean performance by age. Shaded areas represent 95% CIs. The dashed line demarcates chance level and the dots represent individual means. The colored dots and annotation indicate when children's performance exceeds chance level.

Discussion

Analyses established that children master the condition *Absolute Position* at 45 months, making it the easiest task in study 2. Then in quick succession, children succeed in *Orientation of Feature* at 46 months, *Relative Position* at 49 months and *Orientation of Object* at 50 months. Results confirm the hypothesis that children as a group would succeed earlier in *Absolute Position* than in *Relative Position* with a clear 5-months offset. We hypothesized further that children would succeed earlier with *Orientation of Object* than *Orientation of Feature*, but find the opposite to be true. The impression of direction or congruence between cue and target appears to be much more salient for children in graphic stimuli with a salient feature pointing out such as in *Orientation of Feature*, than an overall alignment in the horizontal, vertical or diagonal orientation.

Study 3

The final study features symbol-referent relationships that are based on analogies in size or number without cues and targets sharing similarities in other visual aspects, and while avoiding to draw on conventions in graphic communication. In order to correctly interpret the symbol-referent relationships employed here, children need to infer conceptual order in graphical displays. Such processes can be fostered or obstructed by surface level features of the stimuli at hand which makes it harder to assess whether the age at which children master a task - our primary aim of investigation - depends more on the ability to generally form symbol-referent relationships from analogies or from children's developing ability to process visual complexity. The two base conditions of study three are *Size of Object* and *Size of Number.* Here the cues refer to a target shape as a whole via an analogy in size or number. As a continuous magnitude, size may be easier to grasp and compare for children than discrete concepts (Leibovich, Katzin, Harel, & Henik, 2017). They are complemented by the conditions *Size of Feature* and *Number of Feature* in which the cues refer to a salient aspect of the target stimuli. In these cases, extracting conceptual information is arguably more demanding. Children need to shift attention from a holistic apprehension to exploring specific aspects in isolation, which should make feature tasks more demanding (Callaghan, 1990; Diamond, Carlson, & Beck, 2016; Shepp, Barrett, & Kolbet, 1987). The overall performance across the ages as well as the relative offset in the age at which children solve a task based on symbol-referent relationships targeting objects or their features,

For Review Only

can then serve to evaluate such surface-level effects across two different domains. Prior to data collection, we hypothesized that children will succeed earlier with symbol-referent relationships based in size than in number, and that children will succeed earlier when cues refer to a target object per se rather than a salient feature thereof.

Stimuli

In *Size of Object* the targets in a each trial are identical shapes that are either small or large with regard to the reference frame they are presented in. To make cue and target shapes as distinct as possible they are again employing either squares or circles respectively or fully abstract shapes such as a random scribble or straight lines (cf. figure 4 B). For comparability, *Size of Feature* employs the same cues as *Size of Object*, but features target shapes with either a relatively large or small void, opening, or protrusion. In *Number of Object*, cues and targets are composed of different shapes with some being simple line drawings. The displayed quantities were 1 versus 3, and 2 versus 4. Three- and four-year-olds can reliably distinguish these magnitudes explicitly and implicitly without counting (Mix, Huttenlocher, & Levine, 2002). Special attention was paid to the arrangement of the objects to ensure that the cue and target objects do not share visually similarity by forming a similar pattern, which is difficult as such small number arrays lend themselves to being grouped into canonical shapes by Gestalt principles such as proximity and closure. This issue was addressed by presenting the targets depicting the magnitudes three and four as if outlining irregular shapes. By contrast, the corresponding cues were aligned along a vertical of horizontal axis. For comparability, *Number of Feature* employs the same cues as *Number of Object*. To evoke a sense of quantity in the closed forms serving as choice options, the target shapes in *Number of Feature* either have salient protrusions, or partial areas resulting from incisions. For an overview of all stimuli presented in Study 3, see figure 4.

For Review Only

Participants and Sample

A total of 99 three- to seven-year-old children (M = 59.88 months, SD = 13.44 months, range 36 - 83 months; 55 female) participated. None of these had participated in the previous studies. In addition, 23 children (7 female) were tested but excluded for low performance during familiarization (N = 12), for not completing at least eight out of 16 test trials (N = 1), or being fussy (N = 3). Further exclusions were necessary due to language problems (N = 4) and technical issues (N = 3). For study three, 1559 trials (mean per condition = 389.75, range: 388 - 392) from 99 participants were submitted for analysis.

Results

PPCs indicated excellent fit of observed data and model predictions in both models. Model diagnostics were drawn for the full and null model in study 3. Rhat values in both models were equal to one, indicating convergence across all chains. Effective sample sizes for all fixed effects in the full model (Bulk ESS, mean = 6552, range 4556 - 8586) and the null model (Bulk ESS, mean = 7253, range 5531 - 8885) were > 1000, indicating reliable posterior estimations.

When comparing performance, the full model showed a better fit (ELPD = -941.75) relative to the null model (ELPD = -945.94). The WAIC values also favored the full model (WAIC = 1,883.49) over the null model (WAIC = 1,891.88). Despite the slightly lower WAIC and higher ELPD of the full model, the difference in predictive accuracy (ELPD Diff = -4.19) remains almost within the range of sampling uncertainty (SE = 3.97). In the absence of substantial differences, the full model is reported below in line with the preregistration.

Overall, children's performance increased with age (β = 0.70, 95% CrI [0.44, 0.97]) and with trial number (β = 0.21, 95% CrI [0.06, 0.36]), indicating general improvement across

For Review Only

development and time-on-task. Relative to *Size of Object*, participants were substantially less accurate in *Size of Feature* (beta = -0.71, 95% CrI [-1.06, -0.36]). A smaller, marginal effect was observed in *Number of Feature* (β = -0.31, 95% CrI [-0.67,0.05]). *Number of Object* (β = -0.08, 95% CrI [-0.43, 0.28]) did not differ reliably from *Size of Object*. Age moderated performance less strongly in *Size of Feature* (β = -0.50, 95% CrI [-0.82, -0.20]) and *Number of Feature* (β = -0.29, 95% CrI [-0.60, 0.03]), suggesting lower developmental gains compared to *Size of Object*. Generally, the conditions relying on feature-based reference are associated with lower overall performance and weaker developmental gains. The best overview of the relative performance across conditions and the corresponding age of success is provided by plotting the model estimates (cf. figure 4).

For Review Only

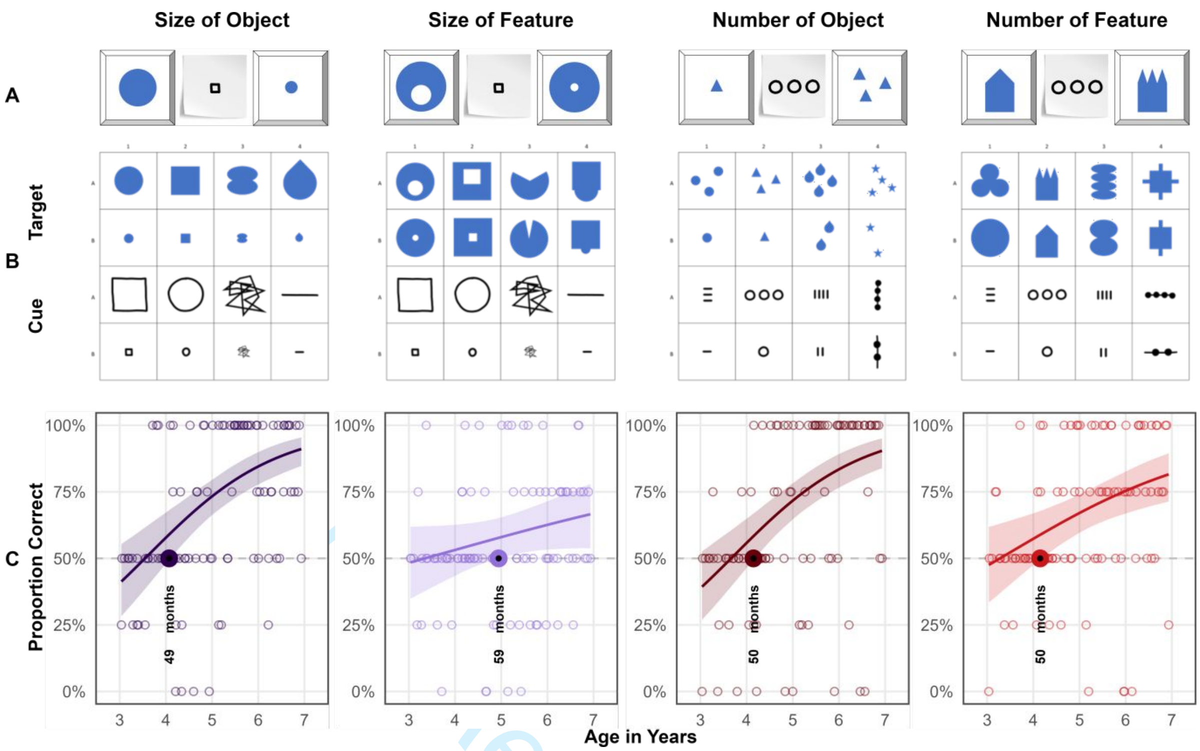

Figure 4. *Stimuli and Developmental Trajectories for Study 3.* Panels illustrate (A) an example stimulus combination (distractor, cue, target), (B) all items of the respective conditions and (C) results. Coloured lines indicate smoothed mean performance by age. Shaded areas represent 95% CIs. The dashed line demarcates chance level and the dots represent individual means. The colored dots and annotation indicate when children's performance exceeds chance level.

Discussion

Model estimates indicate group level success in *Size of Object* at 49 months, *Number of Object* at 50 months, and *Number of Feature* at 50 months. The exception to this pattern is *Size of Feature* where children master the task no sooner than 59 months of age. We hypothesized that children will succeed earlier with analogies in size than in number, and that children would succeed earlier when cues refer to the target objects per se rather than salient features thereof. We find limited support for the hypothesis that the feature conditions are generally more demanding.

That children succeed a month earlier in *Number of Object* than in *Size of Object* as hypothesized is negligible especially in the context of the four-year age-range that is considered here. While *Size of Object* demonstrates when children can grasp size analogies earlier, *Size of Feature* stands out as the most difficult condition in this series of studies. Success here requires shifting attention from general appearance to nested object features and this ability develops considerably throughout the preschool years (Callaghan, 1990; Diamond et al., 2016; Shepp et al., 1987).

For Review Only

Discussion

We presented a concerted three-part research project tracing young children's ability to spontaneously use unfamiliar graphic displays as communicative cues in an object-choice task across a wide variety of symbol-referent relationships. In three studies with four conditions each, we collected data of two children per month of age continuously from the third to the seventh birthday for a final sample of 304 children (Study 1, N = 106; Study 2, N = 99; Study 3, N = 99). We determined the month of age at which children as a group perform above chance with each symbol-referent relationship using Bayesian GLMMs. Across the various conditions, symbolic reference between cue and target was established via direct representation, a part-whole relationship, analogies in size, number and shape, as well as gestalt-like principles like figure-ground relationships, orientation and alignment. Whereas previous work on graphic communication has focused almost exclusively on iconicity as a way of creating meaning and compared performance across binned age-groups, the three studies presented here map development continuously using a highly simple and streamlined procedure across various conceptual dimensions.

By design, the three studies provide a conservative test case of children's ability to find meaning in the symbol-referent relationships presented. Participants are never directly instructed to search specific similarities across cue and target nor prompted to find matches per se as is commonly done in match-to-sample designs (e.g. "Which ones go together?"). Rather children were told that their interlocutor would like to help them find an item in a game of hide and seek and would create a drawing for them. Children then had to infer that the interlocutors drawings may be read as novel symbolic cues referring to either choice option. To prevent learning or demand effects across trials, children did not receive feedback for their responses, only saw one version of the two possible cues for each target-distractor pairing, and were presented with a novel stimulus combination in each trial. Hence, children's inferences depended almost exclusively on the specific graphic arrangement immediately in front of them.

In Study 1, we found that in *Representation* - with cues that are direct outline drawings of the items they refer to - children are robustly better than chance at the lower boundary of our age-window. A second condition, *Pars Pro Toto*, used the same targets as *Representation*, but the cues provided only half of the respective figures to guide children's choices, which children can use at 38 months of age. This demonstrates that by three years of age, children's symbolic competence is still fragile if additional steps such as the straightforward completion of a canonical shape is necessary for referential inference. Two further conditions aimed at abstracting from the cue while still retaining characteristic features of the target shapes with regard to an analogy in form, namely whether they were more round or edgy. *Simple Form Analogy* and *Complex Form Analogy* yield success at 50 months and 47 months. This is the first time a developmental setup features an abstract analogy in shape as a means of disambiguation in graphic communication. Taken together, study 1 provides a within-subjects comparison of young children's ability to read representational and metaphorical uses of form in graphic

For Review Only

communication, showing that while iconicity marks the entry point for the comprehension of graphic cues, more abstract representations are only accessible by the fourth year.

Study 2 explored whether children can infer symbol-referent relationships based on orientation and position. In *Absolute Position*, cue and target placement were aligned in their frame of presentation (e.g. the paper that the symbol was drawn on) and the symbol-referent relation was established by the respective positions in their immediate visual context. Here children succeeded at 44 months of age. In *Relative Position*, target and cue were composed of two figures each that were either closer together or further apart, or aligned on a vertical or horizontal axis. Here children perform above chance at 49 months of age. Success even prior to the fourth birthday in these conditions is remarkable as it requires children to identify and compare relations (*Absolute Position* – figure/ground; *Relative Position* – figure/figure) which is demanding even later throughout the preschool years in other contexts (Christie & Gentner, 2014; Kroupin & Carey, 2022b; Shivaram, Shao, Simms, Hespos, & Gentner, 2023). *Orientation of Object* featured elongated shapes that were oriented either horizontally or vertically while cue and target shared the same direction. Previous research employing a same-different task indicates that children are sensitive to spatial alignment along a common axis in juxtaposed graphic displays no sooner than six to eight years of age (Zheng, Matlen, & Gentner, 2022). In the context of our study, children succeeded at 50 months. In summary, Study 2 addresses an interesting caveat in previous research by presenting stimulus sets where cue and target feature different shapes but are composed in similar ways. While this requires uncovering abstract relations, these relations are by design still based on resemblance on the level of the overall composition, and hence, cover a middle ground between iconic and abstract representations.

For Review Only

Study 3 set out to further reduce iconicity by drawing abstract analogies in size and number. *Number of Object* featured graphic displays with one or three items for target and distractor. *Number of Feature* reuses the same cues but employs targets varying in number with regard to salient features (e.g., spikes or dents). Children succeed in both conditions at 50 months. A different pattern emerges for the two conditions featuring size relationships. Children succeed at 49 months of age in *Size of Object*, but ten months later in *Size of Feature*.

A set of major findings emerges from the studies presented here. First, our results confirm that symbol-referent relationships are most accessible when based on iconicity, becoming increasingly difficult as visual similarity decreases (Ganea, Pickard, & DeLoache, 2008). This progression likely reflects the fact that access to abstract conceptual information matures later in the preschool years (Callaghan, 2000). Furthermore, children are less likely to spontaneously align or compare stimuli when the disparity between comparands increases, a constraint that limits the discovery of non-obvious matches (Gentner & Hoyos, 2017). These findings are of practical relevance for early childhood education as they showcase children's ability to use various expressive dimensions in the graphic domain and can guide the design of pedagogical materials by showing specifically how accessible various ways of conveying meaning in graphic design may be. Second, the reduced accuracy in analogy-based and feature-referenced tasks are both consistent with evidence that younger children tend to focus on surface features rather than relations (Gentner & Hoyos, 2017; Rattermann & Gentner, 1998; Richland et al., 2006), making tasks that require attentional shifting or nested comparison structures particularly demanding (Halford et al., 1998). Shifting attention to different aspects of stimuli sets requires children to keep track of features they have explored. Therefore, relational or abstract as opposed to feature-based processing taxes working memory and inhibitory control more strongly (Gick & Holyoak, 1980; Goswami, 1991). This, in turn, points to aspects of cognitive maturation that are

For Review Only

likely to foster children's performance in the tasks outlined above, as seen in work on the development of executive functions — particularly the early emergence and developmental trajectories of working memory, inhibitory control, and cognitive flexibility in childhood (De Luca & Leventer, 2010; Diamond, 2013; Reilly, Downer, & Grimm, 2022). Fourth, just around their fourth birthday, children are generally able to identify various types of analogies spontaneously in unfamiliar symbolic displays. Previous findings have challenged assumptions of domain specificity in symbol development modalities such as graphic, gestural or verbal communication and rather highlighted the flexibility of symbolic and pragmatic abilities (Callaghan, 1999). Our findings champion a similar perspective with regard to various conceptual or basic graphic dimensions and underline the flexibility of symbolic cognition across various symbol-referent relationships. For conditions that relied strongly on children to abstract from surface level features and focus on analogies in form, orientation, size and number, there is no strong evidence for sequences in the ages of success. Rather, children appear to master all symbol-referent relationships almost at the same point in time, which is highly indicative of a major shift in pragmatic and analogical reasoning skills in the fourth year that drive children's ability for solving communicative coordination problems (Bohn, Kachel, & Tomasello, 2019).

For Review Only

To the extent that the studies presented above highlight the fourth birthday as a milestone in symbolic development, this connects to a rich literature on major developmental shifts in this period of middle-childhood. As change at this age is again rapid and foundational across a variety of cognitive processes (Karmiloff-Smith, 1994; Rakoczy, 2022), several factors are likely to contribute to the convergence of above-chance performance across the experimental conditions used here. What has been described as the four-year-revolution in social cognition endows children with a full-fledged metarepresentational theory of mind and firmly developed belief-desire psychology (Gopnik & Astington, 1988; Rakoczy, 2022; Wellman, Cross, & Watson, 2001). Theory of mind performance, in turn, was shown to be correlated with representational insight into the standing-for-relationship of pictorial representations (Callaghan, Rochat, & Corbit, 2012), as well as insight into communicative intent (Callaghan & Rochat, 2008), and major shifts in perspective-taking abilities (Flavell, Everett, Croft, & Flavell, 1981). All of which are necessary or conducive for the flexible and spontaneous interpretation of graphic displays as well as inference-based or pragmatically challenging communicative behavior broadly construed. Specifically helpful for the tasks employed here may be developmental gains typically occurring around the fourth birthday in the domain of analogical reasoning (Christie & Gentner, 2014; Gentner, 1977; Shivaram et al., 2023), which are again fostered by a suite of executive function skills improving significantly during this important period in child development (Bohn, Tessler, Kordt, Hausmann, & Frank, 2023; Thibaut, French, & Vezneva, 2010). Focusing on various aspects of the stimulus-combinations, working memory (Richland et al., 2006; Simms, Frausel, & Richland, 2018) is crucial for keeping track of features children have explored. To succeed they are furthermore required to suppress irrelevant aspects of the stimuli and inhibit distracting impulses solving relatively abstract tasks (Carlson & Wang, 2007; Jablonski, 2014). Finally, cognitive flexibility is not only key for viewing the same stimulus from different conceptual angles but also specifically for switching modes of interpretation across conditions (Cuevas et al., 2014; Willoughby, Wirth, & Blair, 2012).

While work on analogical reasoning abilities has highlighted that preschoolers are generally able to identify agreement in graphic displays and specifically involving size, number, and shape also at the age of four (Kroupin & Carey, 2022b, 2022a), this has mostly been established in match-to-sample paradigms where children are conventionally presented with three items and prompted with a phrase such as “which ones go together” directly inducing the cognitive operation tested. In our task, children make matches and comparisons across various

For Review Only

dimensions without any verbal scaffolding addressing the match-making process or the dimension of comparison and, hence, showcases abilities for implicit judgement across a wide variety of basic conceptual and visual features. Therefore, our work provides a crucial extension to the literature on analogical reasoning by showing that children can actually put their developing skills to work in the practical context of a versatile communicative coordination game requiring both relational reasoning and pragmatic inference (Cooperrider & Goldin-Meadow, 2017).

The studies described here have limitations that future research needs to address. The combinations of outline drawings and geometric shapes we used here could also be presented in a standard match-to-sample procedure in which children are solely asked to find a match between cue and target. When directly instructed to match stimuli and without the need for considering context, children may succeed earlier. To connect the work presented here with the literature on analogical reasoning, future work should present children with the same set of stimuli in the context of both paradigms. Another general problem is that the project aims to establish robust developmental trajectories but does so only in a single, industrialized population (Henrich, Heine, & Norenzayan, 2010). Future work needs to address the role of enculturation and specifically the influence of permanent exposure to symbolic and pictorial illustrations, writing and numbers or depictions featuring canonical shapes - ideally doing so both by investigating children in various cultural settings, as well as less industrialized communities (Callaghan et al., 2012; Dehaene, Izard, Pica, & Spelke, 2006). Second, our work needs to be complemented with studies employing an individual-differences perspective to untangle the relative contributions of knowledge about various conventional systems of reference, pragmatic skills and the growth of basic level cognitive capacities. Finally, it would be highly valuable to complement the current results on the comprehension of unfamiliar symbols by establishing at what ages children would

For Review Only

be able to use their skills in the production of graphic communication and, thereby, inform theories on the development of communication systems broadly construed (Bohn, Kachel, et al., 2019; Callaghan, 1999; Nölle, Fusaroli, Mills, & Tylén, 2020; Raviv & Arnon, 2018). The results here demonstrate that at least by the age of four, they would be apt receivers of symbolic creations whose reference is based on iconicity, synecdoche, alignment, orientation, as well as analogies in form, size and number.

Taken together, our work provides one of the most comprehensive, robust and systematic investigations of young children's ability to find meaning in unfamiliar graphic displays and may provide a benchmark data set for the development of preschoolers symbolic competence. Its straightforward streamlined design, conceptual breadth, robust sample, and open but fine-grained statistical approach offer a reliable foundation for precisely the future work outlined here and already address an important caveat in our understanding of children's growing aptitude in graphic communication: the conceptual space spanning from iconic to abstract graphical representation.

For Review Only

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For Review Only

SUPPLEMENTARY MATERIAL for DOIXXXXXXXXXXX Young Children's Spontaneous Comprehension of Symbol-Referent Relationships in the Graphic Domain

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SUPPLEMENTARY MATERIAL for DOIXXXXXXXXXX Young Children's Spontaneous Comprehension of Symbol-Referent Relationships in the Graphic Domain

Graphic Overview of Participants and Exclusions

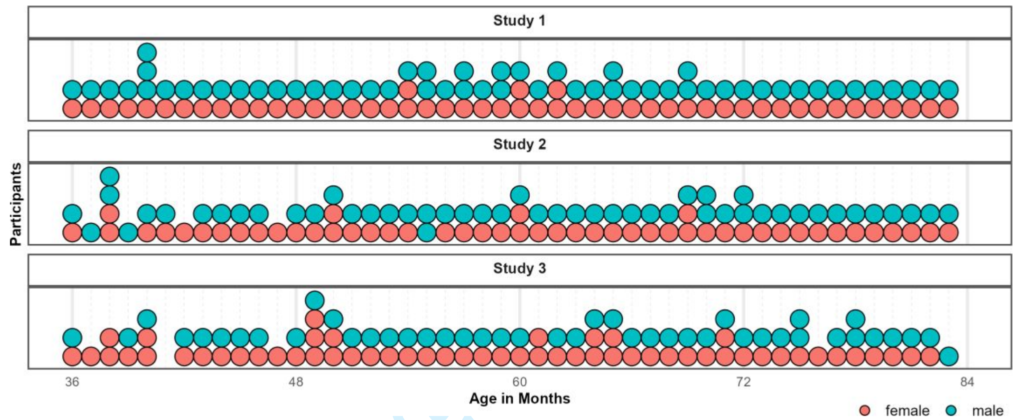

Figure 1. Distribution of Participants across the age-range in all three studies. Dots represent individuals. Colors indicate their respective sex.

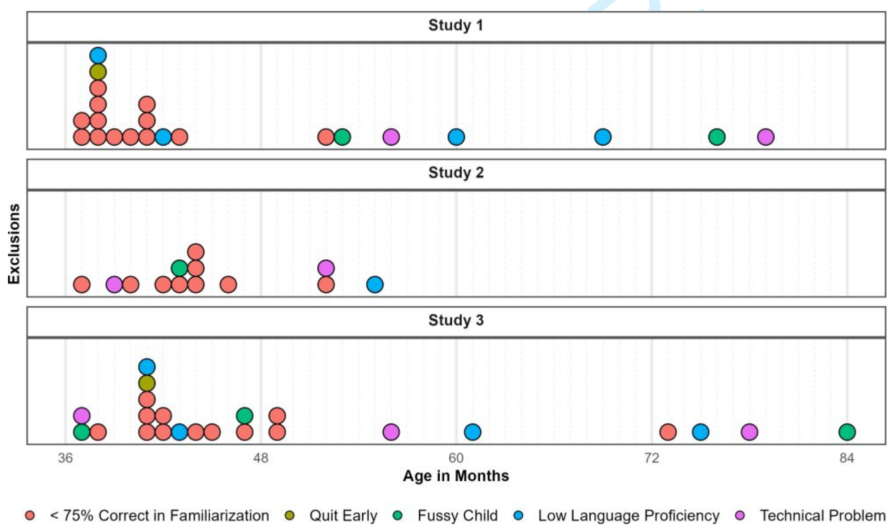

Figure 2. Distribution of Exclusions across the age-range in all three studies. Dots represent individuals. Colors indicate why children where not submitted to analyses.

Posterior Estimates, Descriptives and Conventional Analyses

Study 1

Table 1: Study 1 - Posterior Estimates for the Full Model.

Predictor	Estimate	SD	MAD	95% CrI	Bulk ESS	Tail ESS
Intercept	2.03	0.20	0.20	[1.64, 2.43]	5,184	6,617
Pars Pro Toto	-0.14	0.21	0.21	[-0.54, 0.28]	7,491	7,577
Simple Form Analogy	-1.39	0.18	0.18	[-1.75, -1.03]	6,933	7,821
Complex Form Analogy	-1.36	0.19	0.18	[-1.73, -1.00]	6,810	6,666
Age*	0.42	0.15	0.15	[0.13, 0.72]	4,323	5,827
Trial*	-0.01	0.07	0.07	[-0.15, 0.12]	10,098	7,496
Sex (Male)	-0.29	0.20	0.20	[-0.70, 0.10]	4,512	6,169
Pars Pro Toto × Age*	0.30	0.18	0.18	[-0.05, 0.67]	5,856	7,147
Simple Form Analogy × Age*	-0.20	0.16	0.16	[-0.52, 0.13]	5,344	6,849
Complex Form Analogy × Age*	-0.08	0.17	0.17	[-0.40, 0.25]	5,673	7,198

Note. Estimates represent posterior means with 95% equal-tailed credible intervals (CrIs). MAD indicates Median Absolute Deviation. ESS refers to effective sample size. $\hat{R}$ values omitted as they are all ~1 indicating convergence. * = variables were standardized.

Table 2: Study 1 - Descriptives and Conventional Analyses.

Condition	Age	N	Trials	Trials/N	M	SD	p	df	t(N-1)	d
Representation	3-year-olds	26	103	3.96	75.00	22.36	<0.001	25.00	5.70	1.12
	4-year-olds	28	109	3.89	81.25	33.07	<0.001	27.00	5.00	0.94
	5-year-olds	28	112	4.00	87.50	28.46	<0.001	27.00	6.97	1.32
	6-year-olds	24	96	4.00	87.50	20.85	<0.001	23.00	8.81	1.80
Pars Pro Toto	3-year-olds	26	104	4.00	66.35	22.30	<0.001	25.00	3.74	0.73
	4-year-olds	28	112	4.00	75.00	31.91	<0.001	27.00	4.15	0.78
	5-year-olds	28	112	4.00	85.71	21.97	<0.001	27.00	8.60	1.63
	6-year-olds	24	96	4.00	91.67	15.93	<0.001	23.00	12.82	2.62
Simple Form Analogy	3-year-olds	26	104	4.00	53.85	23.12	0.404	25.00	0.85	0.17
	4-year-olds	28	112	4.00	59.82	28.33	0.078	27.00	1.83	0.35
	5-year-olds	28	112	4.00	58.93	28.23	0.106	27.00	1.67	0.32
	6-year-olds	24	96	4.00	69.79	23.29	<0.001	23.00	4.16	0.85
Complex Form Analogy	3-year-olds	25	100	4.00	53.00	25.33	0.559	24.00	0.59	0.12
	4-year-olds	28	112	4.00	54.46	23.62	0.326	27.00	1.00	0.19
	5-year-olds	28	112	4.00	59.82	25.77	0.054	27.00	2.02	0.38
	6-year-olds	24	96	4.00	76.04	18.77	<0.001	23.00	6.80	1.39

Note. Descriptive and inferential statistics by age group and condition. Performance (*M*, *SD*) is percent correct responses.*p*, *t*, and *d* reflect one-sample t-tests vs. chance level (0.5). *Trials* = total trials across participants; *Trials/N* = average per participant.

Study 2

Table 3: Study 2 - Posterior Estimates for the Full Model.

Predictor	Estimate	SD	MAD	95% CrI	Bulk ESS	Tail ESS
Intercept	1.08	0.20	0.20	[0.69, 1.48]	3,776	6,041
Relative Position	-0.28	0.18	0.18	[-0.64, 0.08]	7,227	7,370
Orientation of Object	-0.10	0.19	0.19	[-0.46, 0.27]	7,545	7,408
Orientation of Feature	-0.12	0.18	0.18	[-0.48, 0.23]	7,402	7,559
Age*	0.40	0.14	0.14	[0.14, 0.68]	4,012	4,964
Trial*	0.26	0.08	0.08	[0.10, 0.43]	5,662	7,109
Sex (Male)	-0.01	0.22	0.22	[-0.45, 0.43]	3,485	5,359
Relative Position × Age*	-0.06	0.16	0.16	[-0.37, 0.25]	7,445	7,754
Orientation of Object × Age*	0.19	0.16	0.16	[-0.12, 0.51]	7,513	7,844
Orientation of Feature × Age*	-0.01	0.16	0.16	[-0.32, 0.31]	7,428	8,385

Note. Estimates represent posterior means with 95% equal-tailed credible intervals (CrIs). MAD indicates Median Absolute Deviation. ESS refers to effective sample size. $\hat{R}$ values omitted as they are all ~1 indicating convergence. * = variables were standardized.

Table 4: Study 2 - Descriptives and Conventional Analyses.

Condition	Age	N	Trials	Trials/N	M	SD	p	df	t(N-1)	d
Absolute Position	3-year-olds	20	80	4.00	58.75	18.63	0.049	19.00	2.10	0.47
	4-year-olds	25	100	4.00	62.00	33.17	0.083	24.00	1.81	0.36
	5-year-olds	27	108	4.00	73.15	30.95	<0.001	26.00	3.89	0.75
	6-year-olds	25	100	4.00	81.00	29.12	<0.001	24.00	5.32	1.06
Relative Position	3-year-olds	22	88	4.00	54.55	21.32	0.329	21.00	1.00	0.21
	4-year-olds	24	96	4.00	59.38	26.39	0.095	23.00	1.74	0.36
	5-year-olds	27	108	4.00	68.52	34.39	0.010	26.00	2.80	0.54
	6-year-olds	25	100	4.00	76.00	34.97	0.001	24.00	3.72	0.74
Orientation of Object	3-year-olds	22	85	3.86	48.86	19.64	0.789	21.00	-0.27	0.06
	4-year-olds	25	100	4.00	57.00	25.54	0.183	24.00	1.37	0.27
	5-year-olds	27	108	4.00	75.93	24.50	<0.001	26.00	5.50	1.06
	6-year-olds	25	100	4.00	81.00	25.29	<0.001	24.00	6.13	1.23
Orientation of Feature	3-year-olds	21	80	3.81	55.95	28.40	0.348	20.00	0.96	0.21
	4-year-olds	25	100	4.00	67.00	28.61	0.007	24.00	2.97	0.59
	5-year-olds	27	108	4.00	65.74	34.77	0.027	26.00	2.35	0.45
	6-year-olds	25	100	4.00	79.00	33.60	<0.001	24.00	4.32	0.86

Note. Descriptive and inferential statistics by age group and condition. Performance (*M*, *SD*) is percent correct responses.*p*, *t*, and *d* reflect one-sample t-tests vs. chance level (0.5). *Trials* = total trials across participants; *Trials/N* = average per participant.

Study 3

Table 5: Study 3 - Posterior Estimates for the Full Model.

Predictor	Estimate	SD	MAD	95% CrI	Bulk ESS	Tail ESS
Intercept	1.04	0.18	0.18	[0.70, 1.38]	5,074	6,931
Size of Feature	-0.71	0.18	0.18	[-1.06, -0.36]	7,351	7,601
Number of Object	-0.08	0.18	0.18	[-0.43, 0.28]	7,735	8,265
Number of Feature	-0.31	0.18	0.18	[-0.67, 0.05]	7,429	7,348
Age*	0.70	0.14	0.13	[0.44, 0.97]	4,556	6,771
Trial*	0.21	0.08	0.08	[0.06, 0.36]	8,586	7,671
Sex (Male)	-0.18	0.19	0.18	[-0.55, 0.18]	5,392	6,583
Size of Feature × Age*	-0.50	0.16	0.16	[-0.82, -0.20]	6,020	6,979
Number of Object × Age*	0.00	0.16	0.16	[-0.32, 0.32]	6,741	7,976
Number of Feature × Age*	-0.29	0.16	0.16	[-0.60, 0.03]	6,633	6,930

Note. Estimates represent posterior means with 95% equal-tailed credible intervals (CrIs). MAD indicates Median Absolute Deviation. ESS refers to effective sample size. $\hat{R}$ values omitted as they are all ~1 indicating convergence. * = variables were standardized.

Table 6: Study 3 - Descriptives and Conventional Analyses.

Condition	Age	N	Trials	Trials/N	M	SD	p	df	t(N-1)	d
Size of Object	3-year-olds	21	84	4.00	51.19	23.02	0.815	20.00	0.24	0.05
	4-year-olds	27	105	3.89	54.63	31.80	0.456	26.00	0.76	0.15
	5-year-olds	26	104	4.00	84.62	25.57	<0.001	25.00	6.90	1.35
	6-year-olds	24	96	4.00	78.12	22.50	<0.001	23.00	6.12	1.25
Size of Feature	3-year-olds	21	84	4.00	46.43	19.82	0.419	20.00	-0.83	0.18
	4-year-olds	26	104	4.00	51.92	23.37	0.678	25.00	0.42	0.08
	5-year-olds	26	104	4.00	56.73	32.06	0.295	25.00	1.07	0.21
	6-year-olds	24	96	4.00	62.50	19.50	0.005	23.00	3.14	0.64
Number of Object	3-year-olds	21	84	4.00	42.86	21.13	0.137	20.00	-1.55	0.34
	4-year-olds	27	108	4.00	61.11	30.49	0.069	26.00	1.89	0.36
	5-year-olds	26	104	4.00	72.12	34.88	0.003	25.00	3.23	0.63
	6-year-olds	24	96	4.00	83.33	32.69	<0.001	23.00	4.99	1.02
Number of Feature	3-year-olds	21	82	3.90	50.00	19.36	>0.999	20.00	0.00	0.00
	4-year-olds	27	108	4.00	62.96	24.39	0.010	26.00	2.76	0.53
	5-year-olds	26	104	4.00	65.38	31.68	0.020	25.00	2.48	0.49
	6-year-olds	24	96	4.00	76.04	27.07	<0.001	23.00	4.71	0.96

Note. Descriptive and inferential statistics by age group and condition. Performance (*M*, *SD*) is percent correct responses.*p*, *t*, and *d* reflect one-sample t-tests vs. chance level (0.5). *Trials* = total trials across participants; *Trials/N* = average per participant.

Exploratory Analyses Including Item-Level Random Effects

In addition to our main analyses, we preregistered additional exploratory analyses for evaluating item-level effects separately in each study. For this we added a crossed random effects for items, allowing the relationship between age and performance to vary across items (correct ~ condition*z.age + z.trial + sex + (z.trial |subid)). Due to the lower number of individual items within a task, we expected these models to be less diagnostic with regard to our main research question. However, the analyses provides detail about the equivalence of items within conditions.

For Review Only

Study 1

An exploratory model with additional item-level random effects showed that the variability of intercepts at the item level was small (β = 0.17, 95% CrI [0.01,0.40]). Likewise there was only modest variation for the effect of age across items (β = 0.14, 95% CrI [0.01,0.34]). The correlation between item difficulty and item-specific age slopes was highly uncertain (ρ = 0.35, 95% CrI [-0.83,0.98]). Model weights based on the WAIC suggested that the full model was more likely to provide accurate predictions (weight = 82.92%) compared to the item model (weight = 17.08%).The estimated difference in expected log predictive density (elpd) between models, however, was not much larger than the difference itself (ELPD diff = -0.91), indicating considerable uncertainty as to whether either model was preferable. For on overview of logg odds for all items across conditions see figure 3. For a full overview of coefficients and the random effects structure of the item model see table 7.

Study 2

An exploratory model including item-level random effects indicated modest variability (β = 0.25, 95% CrI [0.02,0.53]). The variability of age effects across items was small (β = 0.11, 95% CrI [0.00, 0.30]), suggesting that the effect of age was fairly consistent across items. The correlation between item intercepts and age slopes was near zero and highly uncertain (ρ = 0.01, 95% CrI [-0.94,0.94]), providing no reliable evidence that slightly easier or difficult items varied in age-related gains. Comparing the models using WAIC weights showed that including item-level random effects in Study 2 did not provide a clear advantage in out-of-sample predictive accuracy (full model 39.32%; item model 60.68%). Comparing the models corroborates this (ELPD WAIC; full model = -889.99; item model = -889.28). The difference in expected log predictive density was negligible (ELPD diff = 0), and within the range of sampling uncertainty (SE =0). For an overview of item level variability across conditions see figure 3. For a full table of coefficients and random effects of the exploratory model see table 8.

For Review Only

Study 3

To evaluate the equivalence of items within conditions, item-level random effects were included in an exploratory model showing that item variation at the intercept was moderate (β = 0.20, 95% CrI [0.02,0.44]) and that the effect of age varied slightly across items (β = 0.22, 95% CrI [0.03,0.45]). Again, the correlation between item difficulty and item-specific age slopes was moderately positive but uncertain (ρ = 0.55, 95% CrI [-0.57,0.99]) providing weak evidence that easier items had stronger age-related gains. Comparing the models using WAIC weights showed that the item model had higher predictive power (full model 31.70%; item model 68.30%). Comparing the models' WAIC via ELPDs corroborates this (full model = -941.75; item model = -939.71). The difference in expected log predictive density favored the item model (ELPD diff = -2.04), though the uncertainty around this estimate suggests that the evidence is not decisive (SE =3.19). Graphically exploring item-level variability via logg odds indicates that the estimated deviation from the intercept is still including zero (cf. figure 3). For an overview of coefficients and random effects see table 9 in appendix 3.

Discussion

In studies 1 and 2, the inclusion of item-level random effects did not meaningfully improve model fit. The main analyses reported in the manuscript and the exploratory models reported here account for the data equally well. For study 3, including item-level random effects yielded moderate variability. Here the analysis provides modest evidence for variability due to item level effects. As in all studies reported above, these were small compared to the participant random effects. None of the model comparisons provided substantial evidence for a better fit of the exploratory models. Hence, the additional analyses provided here provide no grounds for deterring from the preregistered analysis reported in the manuscript.

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Table 7: Study 1 - Posterior Estimates for Exploratory Model with Item-Level Random Effects.

Predictor	Estimate	SD	MAD	95% CrI	Bulk ESS	Tail ESS
Intercept	2.04	0.23	0.22	[1.60, 2.50]	6,827	6,155
Pars Pro Toto	-0.13	0.25	0.24	[-0.63, 0.36]	9,329	7,194
Simple Form Analogy	-1.40	0.23	0.23	[-1.85, -0.94]	8,693	7,089
Complex Form Analogy	-1.36	0.23	0.22	[-1.82, -0.91]	8,770	7,373
Age*	0.43	0.17	0.17	[0.09, 0.77]	6,101	5,738
Trial*	-0.01	0.07	0.07	[-0.15, 0.12]	12,068	8,047
Sex (Male)	-0.29	0.21	0.20	[-0.70, 0.11]	7,184	7,229
Pars Pro Toto × Age*	0.31	0.22	0.21	[-0.12, 0.73]	8,061	6,743
Simple Form Analogy × Age*	-0.20	0.20	0.19	[-0.60, 0.19]	7,048	6,608
Complex Form Analogy × Age*	-0.07	0.20	0.19	[-0.48, 0.32]	7,479	6,036
Random Effects						
Item Intercept (SD)	0.17	0.10	0.10	[0.01, 0.40]	3,324	5,071
Item × Age Slope (SD)	0.14	0.09	0.09	[0.01, 0.34]	4,348	5,360
Subject Intercept (SD)	0.85	0.10	0.10	[0.66, 1.06]	4,038	6,548
Subject × Trial slope (SD)	0.17	0.10	0.11	[0.01, 0.37]	4,163	5,072
Correlation: Item Intercept & Age Slope	0.35	0.52	0.53	[-0.83, 0.98]	6,312	7,115
Correlation: Subject Intercept & Trial Slope	-0.48	0.42	0.37	[-0.98, 0.62]	9,528	6,694

Note. Estimates represent posterior means with 95% equal-tailed credible intervals (CrIs). MAD indicates Median Absolute Deviation. ESS refers to effective sample size. $\hat{R}$ values omitted as they are all ~1 indicating convergence. * = variables were standardized.

Table 8: Study 2 - Posterior Estimates for Exploratory Model with Item-Level Random Effects.

Predictor	Estimate	SD	MAD	95% CrI	Bulk ESS	Tail ESS
Intercept	1.10	0.24	0.24	[0.63, 1.58]	6,790	6,800
Relative Position	-0.29	0.27	0.25	[-0.82, 0.24]	8,525	6,333
Orientation of Object	-0.10	0.27	0.25	[-0.63, 0.45]	8,989	6,423
Orientation of Feature	-0.13	0.27	0.25	[-0.68, 0.42]	8,951	6,857
Age*	0.41	0.15	0.15	[0.11, 0.72]	7,413	7,361
Trial*	0.26	0.08	0.08	[0.10, 0.43]	9,367	8,404
Sex (Male)	-0.01	0.23	0.23	[-0.46, 0.44]	6,139	6,947
Relative Position × Age*	-0.06	0.19	0.18	[-0.43, 0.31]	9,354	7,529
Orientation of Object × Age*	0.19	0.19	0.18	[-0.16, 0.56]	9,640	7,368
Orientation of Feature × Age*	-0.01	0.19	0.18	[-0.37, 0.36]	9,415	7,527
Random Effects						
Item Intercept (SD)	0.25	0.13	0.12	[0.02, 0.53]	2,919	4,096
Item × Age Slope (SD)	0.11	0.08	0.08	[0.00, 0.30]	4,974	6,461
Subject Intercept (SD)	0.99	0.12	0.12	[0.76, 1.25]	3,605	5,886
Subject × Trial slope (SD)	0.48	0.12	0.12	[0.25, 0.71]	3,534	4,754
Correlation: Item Intercept & Age Slope	0.01	0.55	0.69	[-0.94, 0.94]	12,775	7,849
Correlation: Subject Intercept & Trial Slope	0.64	0.19	0.19	[0.20, 0.95]	3,746	3,362

Note. Estimates represent posterior means with 95% equal-tailed credible intervals (CrIs). MAD indicates Median Absolute Deviation. ESS refers to effective sample size. $\hat{R}$ values omitted as they are all ~1 indicating convergence. * = variables were standardized.

Table 9: Study 3 - Posterior Estimates for Exploratory Model with Item-Level Random Effects.

Predictor	Estimate	SD	MAD	95% CrI	Bulk ESS	Tail ESS
Intercept	1.06	0.21	0.20	[0.66, 1.48]	7,161	5,863
Size of Feature	-0.71	0.24	0.23	[-1.20, -0.22]	7,855	7,019
Number of Object	-0.08	0.25	0.24	[-0.56, 0.42]	7,859	7,461
Number of Feature	-0.31	0.25	0.23	[-0.81, 0.19]	8,119	6,635
Age*	0.71	0.19	0.18	[0.34, 1.09]	6,586	6,280
Trial*	0.21	0.08	0.08	[0.06, 0.37]	12,182	7,945
Sex (Male)	-0.18	0.19	0.19	[-0.57, 0.20]	8,353	7,734
Size of Feature × Age*	-0.50	0.24	0.23	[-0.98, -0.03]	7,251	6,897
Number of Object × Age*	0.01	0.24	0.23	[-0.47, 0.49]	6,793	6,483
Number of Feature × Age*	-0.28	0.24	0.23	[-0.76, 0.21]	7,445	7,168
Random Effects						
Item Intercept (SD)	0.20	0.11	0.10	[0.02, 0.44]	3,252	3,250
Item × Age Slope (SD)	0.22	0.10	0.09	[0.03, 0.45]	3,196	3,365
Subject Intercept (SD)	0.73	0.10	0.10	[0.54, 0.94]	4,031	5,630
Subject × Trial slope (SD)	0.42	0.11	0.10	[0.18, 0.62]	2,907	2,870
Correlation: Item Intercept & Age Slope	0.55	0.41	0.33	[-0.57, 0.99]	4,265	4,116
Correlation: Subject Intercept & Trial Slope	0.16	0.26	0.26	[-0.36, 0.66]	4,836	4,585

Note. Estimates represent posterior means with 95% equal-tailed credible intervals (CrIs). MAD indicates Median Absolute Deviation. ESS refers to effective sample size. $\hat{R}$ values omitted as they are all ~1 indicating convergence. * = variables were standardized.

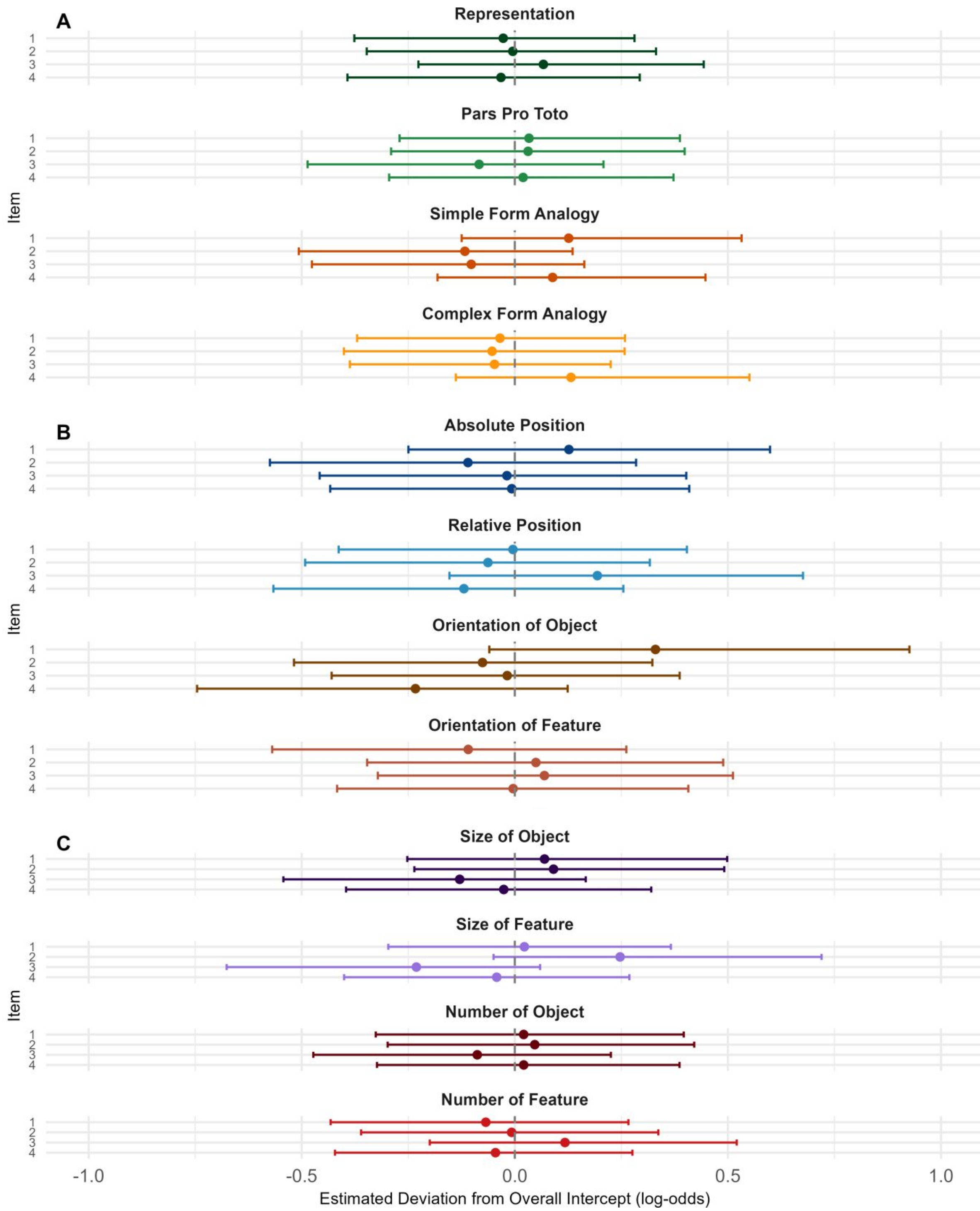

Figure 3. *Item-Level Random Intercepts by Condition.*